


# climate risk in vulnerable mountain regions of the world

## Day 1 (22 May)

Jakob Steiner

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fidelsteiner.github.io

# Global Awareness Education (GAE)

## Transdisciplinary Course Program and Career Service – TRACS
## University of Tübingen



@Angharad Dean

- Courses on global issues
- Globalization, human rights, indigenous knowledge, climate crisis exit, etc.
- International team of experts
- Global Awareness Certificate for 15 ECTS

Read more:
https://uni-tuebingen.de/en/215290

# CIVIS

## Europe’s Civic University Alliance




- Alliance of 11 European universities
- GAE participates in many CIVIS projects!

- In this course, there are students who are registered in the CIVIS micro-program Global Awareness!

Read more:
https://civis.eu/en

Co-funded by the Erasmus+ Programme of the European Union

# Global Awareness Education in CIVIS




Co-funded by the Erasmus+ Programme of the European Union

- CIVIS Micro-Program Global Awareness
  - U of Tübingen, U of Bucharest, and Universidad Autónoma de Madrid
  - 15 ECTS for Bachelor students from all fields, creditable as key qualifications for students from U of Tübingen.

- Blended Intensive Programs (BIP)
  - Online sessions combined with a short trip for in-person teaching

# Course content

**22 May (1400 – 1700)**

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

**27 May (0900 – 1500)**

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

**28 May (0900 – 1500)**

Science to policy

Serious game

**29 May (0900 – 1500)**

Serious game – course wrap up

# Course content

**22 May (1400 – 1700)**

- 14:00 - 14:30 Introduction to the course and topics
- 14:30 - 15:15 Mountains and Risk
- 15:15 - 15:35 Break
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# Self evaluation

How familiar are you with the course topic (climate related risks and/or mountain environments)?

How confident are you in explaining to someone what constitute risks in mountain regions and how they are assessed?

How relevant is this topic in your field of study?

# Course aims

At the end of the course, you will be able to assess

(i) what is 'risk' in a climate change context, what terms do we use, and what do we know and know less about.

(ii) what are ways to assess risks and to mitigate and adapt to them.

(iii) how risks are managed (or mismanaged) at the global stage.

You should also be able to critically reflect on how interdisciplinarity and 'thinking outside the box' can be crucial in risk assessments and how you can shape future approaches with critical reflections.

# What to expect from the course

- to understand what crucial risks in mountain regions are, how we assess them and what attempts to respond to them look like

- to learn how state and non-state actors as well as academics respond to risks and how they are addressed from local all the way to the global scale

# What to bring to the course

- personal reflection on knowledge from your own background

- challenge the status quo – this isn't a topic that is set in stone – your opinion and knowledge matters and your contribution can shape the discourse!

# (academic) background

***BSc*** *(ETH Zurich, Switzerland): Water for agriculture in Xinjiang/Western China , Prof. Wolfgang Kinzelbach*

***MSc*** *(ETH Zurich, Switzerland): Ice cliffs on debris-covered glaciers in Nepal , Prof. Paulo Burlando*

***Concert Diploma Violoncello*** *(Austria/Germany), Prof. Michael Hell*

***PhD*** *(Univ. Utrecht, Netherlands): Debris-covered glaciers in the high mountain water cycle Prof. Walter Immerzeel, Prof. Marc Bierkens*

***Engineer*** *at hydrosolutions ltd. (Switzerland)*

- *agricultural water management in Central Asia, groundwater monitoring in Eastern and Central China (SDC)*

***Glacier-Hydrologist*** *at ICIMOD (Nepal, Pakistan)*

- *group lead of the cryosphere group within the Water and Air Theme , fieldwork and research (Nepal, Pakistan, Afghanistan), trainings for regional member countries, coordination of multi-stakeholder research and policy projects (Nepal, Pakistan, Afghanistan)*

# (academic) background – current positions



***PostDoc at University of Graz (Austria)***

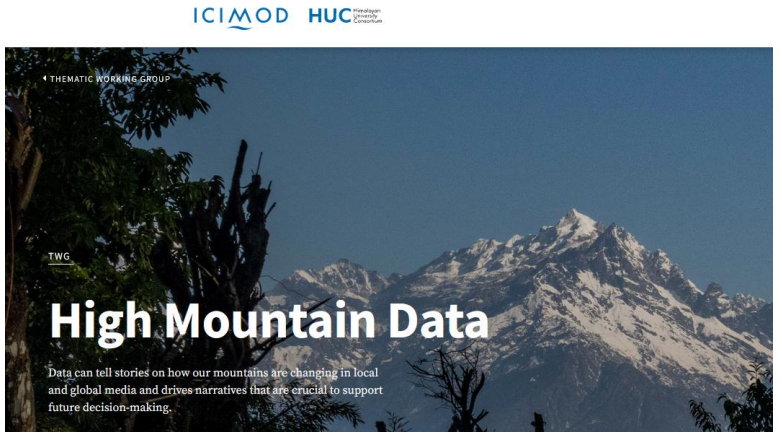
*– research on the Greenland ice margin*

*LATTICE - https://www.uibk.ac.at/en/acinn/research/ice-and-climate/projects/lattice/*

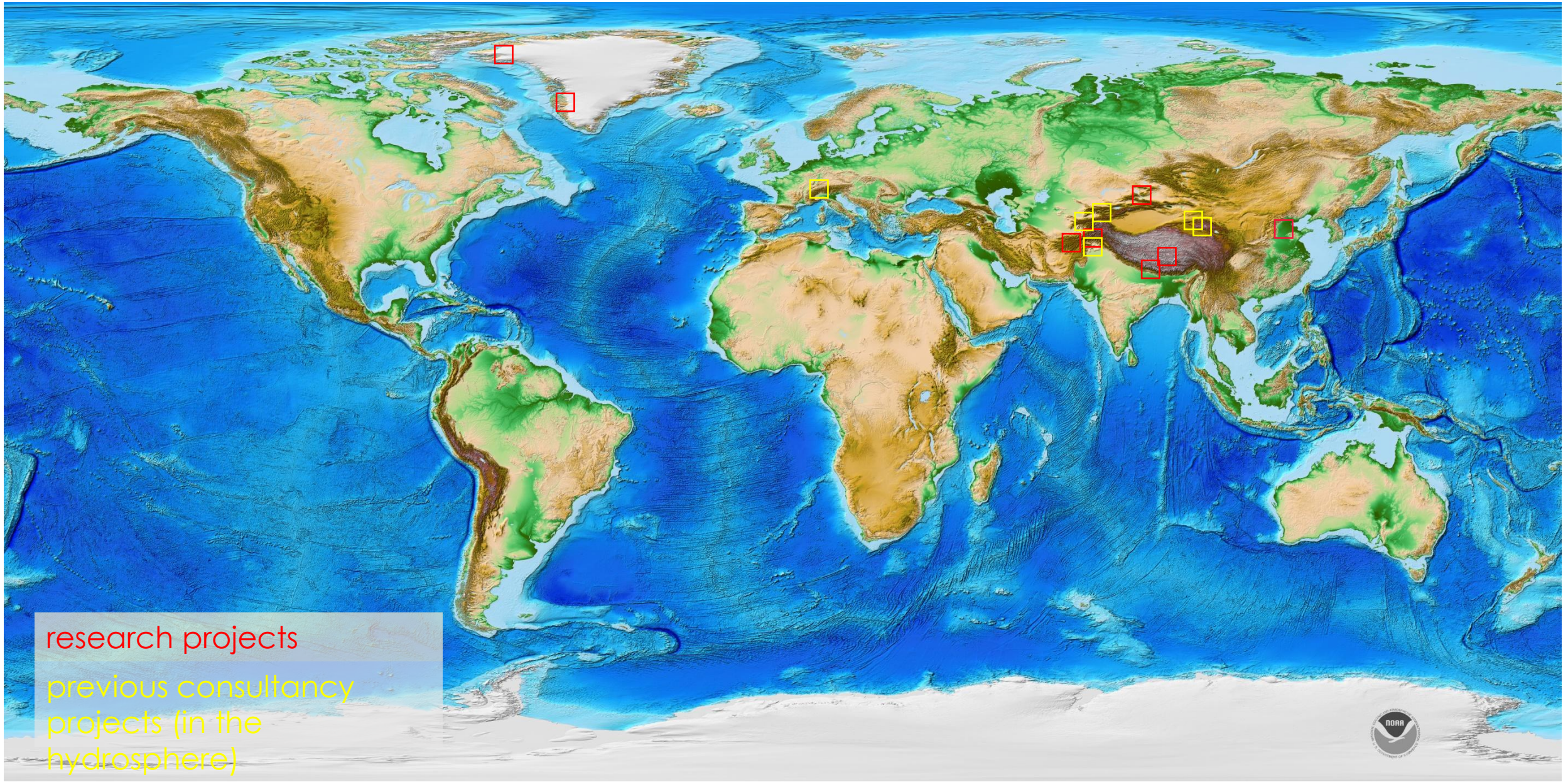
***Fellow at the Himalayan University Consortium (Nepal/Pakistan)*** *– (co)lead of working groups on cryosphere, society and data*

*HUC - https://huc-hkh.org/thematic-working-groups/high-mountain-data*

*HiRISK project lead - https://hirisk.org/*




https://fidelsteiner.github.io/


research projects
previous consultancy projects (in the hydrosphere)

# interdisciplinary transgressions

**Socio-ecological dynamics of diverse global permafrost-agroecosystems under environmental change**

Melissa Ward Jones[1,2], Joachim Otto Habeck[3], Mathias Ulrich[4], Susan Crate[5], Glenna Gannon[2], Tobias Schwoerer[2], Benjamin Jones[2], Mikhail Kanevskiy[2], Prashant Baral[6], Amina Maharjan[6], Jakob Steiner[6,7], Andrew Spring[8], Mindy Jewell Price[9], David Bysouth[10], Bruce C. Forbes[11], Mariana Verdonen[12], Timo Kumpula[12], Jens Strauss[13], Torben Windirsch[13], Christopher Poeplau[14], Yuri Shur[2], Benjamin Gaglioti[2], Nicholas Parlato[2], Fulu Tao[15], Merritt Turetsky[16], Stephanie Grand[17], Adrian Unc[18,19], and Nils Borchard[20]

[1]Corresponing author

Email: mkwardjones@alaska.edu

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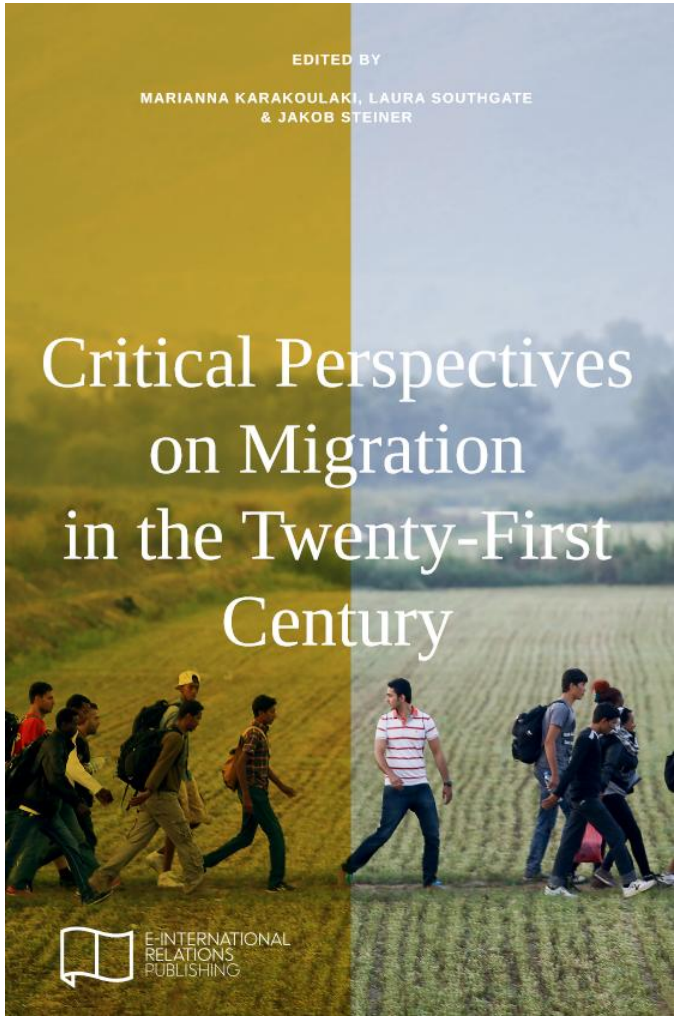
[2]University of Alaska Fairbanks, Fairbanks, USA; [3]Universität Hamburg, Hamburg, Germany; [4]German Environment Agency, Dessau-Roßlau, Germany; [5]George Mason University, Fairfax, USA; [6]ICIMOD, International Centre for Integrated Mountain Development, Lalitpur, Nepal; [7]Department of Geography and Regional Science, University of Graz, Heinrichstraße 36, 8010 Graz, Austria; [8]Wilfrid Laurier University, Waterloo, Canada; [9]University of California - Berkeley, Berkeley, USA; [10]University of Guelph, Guelph, Canada; [11]University of Lapland, Rovaniemi, Finland; [12]University of Eastern Finland, Joensuu, Finland; [13]Alfred Wegener Institute Helmholtz Centre for Polar and Marine Research, Potsdam, Germany; [14]Thünen Institute of Climate-Smart Agriculture, Braunschweig, Germany; [15]Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, China; [16]University of Colorado Boulder, Boulder, USA; [17]Université de Lausanne, Lausanne, Switzerland; [18]Memorial University Grenfell Campus, Corner Brook, Canada; [19]McGill University, St-Anne-de-Bellevue, Canada; and [20]German Agricultural Society, Frankfurt, Germany and University of Lisbon, Lisbon, Portugal

JAKOB STEINER

## NEGLECTED LAND IN BETWEEN – THE OBLIVIOUSNESS TO SMALL TOWN PAKISTAN IN THE WESTERN NARRATIVE

Abstract

The Western imagination of human settlement of the Subcontinent has long been limited to two extreme possibilities - the backward but romantic village in the remote countryside versus the bustling megacity. Ideally, as the representation of the 'modern' Pakistan, India or Bangladesh these worlds collide in literary, cinematic or even scholarly representation. The world in between, life in the small town, the urban peripheries of larger conglomerations and the exchanges between these spaces and their influence on urban growth and rural development have been largely neglected. Not only has this resulted in a very biased imagination of these countries but also in inadequate approaches in infrastructure development funded and thought up outside the country and by the NGO sector that often tries to emulate the western imagination. This paper examines some of these deficiencies. It begins with discussing briefly how Pakistan is framed by the foreign eye that had a chance to actually see the place and where these frames likely originate. It is the simple example of travelling between Lahore and Islamabad, a route often taken by decision makers in international organisations and policy wonks that holds as an initial introduction to the dilemma. The route mostly taken, the Motorway, reproduces the bimodal view of Pakistan of megacity vs. rural village. On the other hand, the GT road, which leads one through the whole spectrum of urban settlements in the country, from the historical metropolis Lahore through the largely unmentioned cities in narratives on Pakistan such as, Gujranwala, Jhelum or Wazirabad to the twin cities, with many smaller towns on the way, is hardly known.




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# Course setup

1. The basics on risk and mountains
2. Current state of the art in research and response
3. Critical reflection on where we are heading
4. You applying your knowledge in a serious game – you turn into a national actor addressing climate risks at a fictional transboundary standoff
5. Wrap up - your own (interdisciplinary) reflections

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# General definitions

Mountains

Risk (Hazard – Vulnerability – Exposure)

Loss and Damage

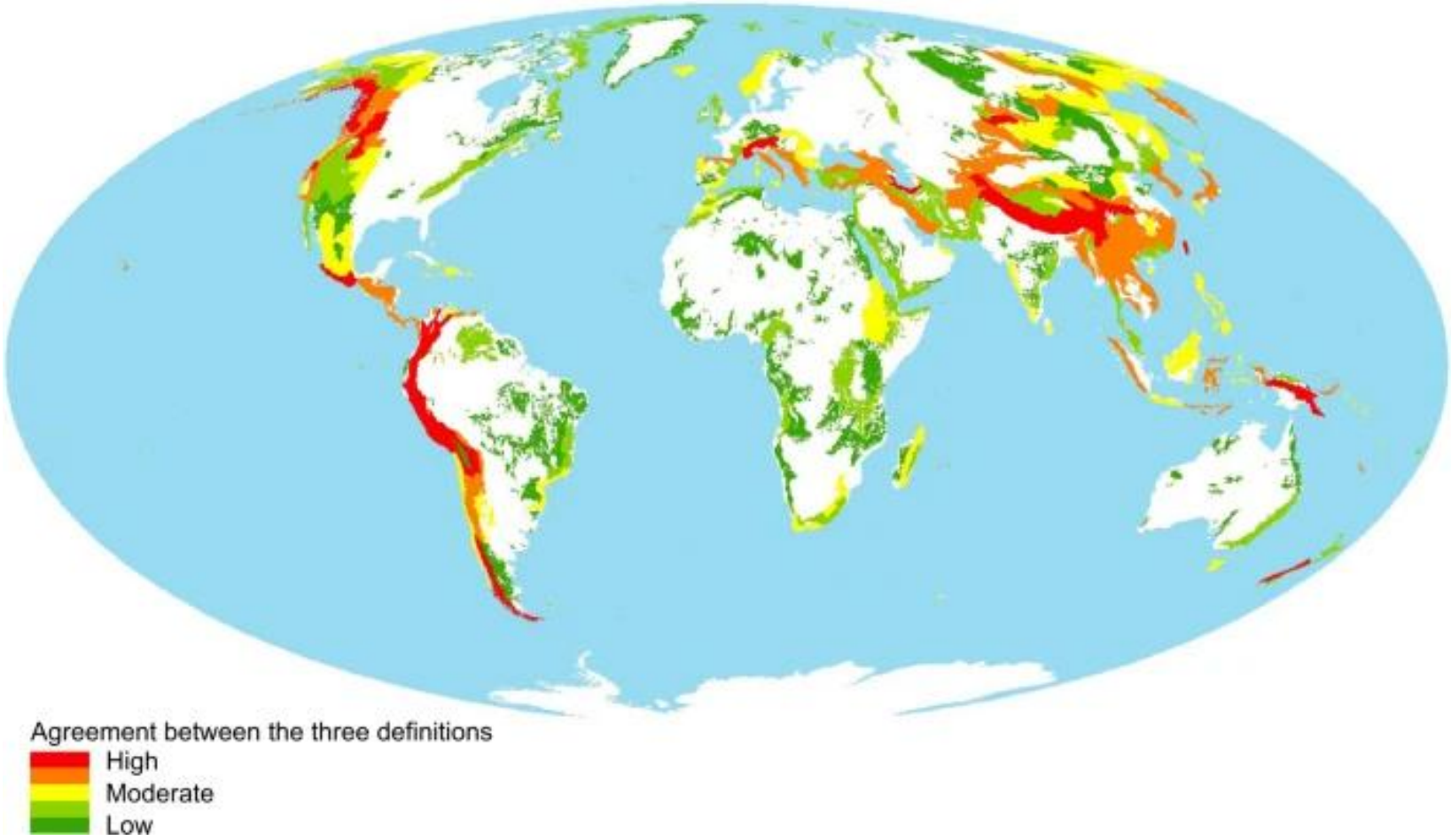
Transboundary

# General definitions

***mountains*:**

not that easy to agree on where they are

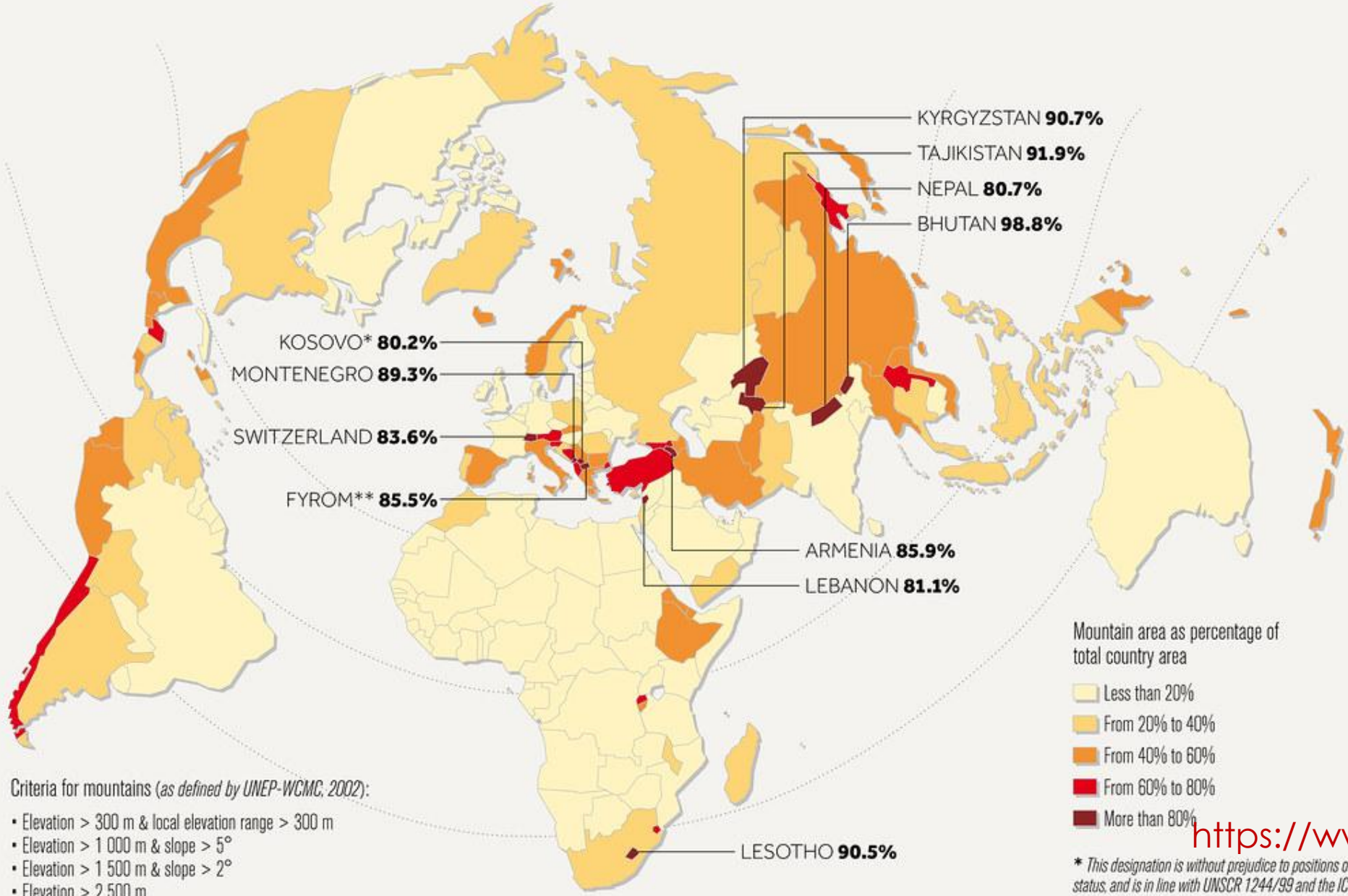
large areas downstream impacted by mountains (melt water, sediment supply, migration …)




Snethlage et al. (2022)
https://www.earthenv.org/mountains


DÄNEMARK
NORDSEE
OSTSEE
Deutsche Bucht
Nordfriesische Inseln
Ostfriesische Inseln
Helgoländer Bucht
Arkonasee
Kieler Bucht
Mecklenburger Bucht
Lübecker Bucht
Pommersche Bucht
Rügen
Usedom
Vorpommern
Fehmarn
Kiel
Flensburg
Sylt
Nordfriesland
Bungsberg 167 m
Stralsund
Rostock
Greifswald
Lübeck
Wismar
Schwerin
Neubrandenburg
Helpter Berge 179 m
Mecklenburgische Seenplatte
Uckermark
Wilhelmshaven
Ostfriesland
Emden
Bremerhaven
HAMBURG
Lüneburg
Oldenburg (Oldenburg)
Bremen
Lüneburger Heide
Tiefland
Altmark
Havelland
POLEN
NIEDER-LANDE
Norddeutsches
Emsland
Lingen (Ems)
Wolfsburg
Hannover
Braunschweig
Brandenburg an der Havel
Potsdam
BERLIN
Frankfurt (Oder)
Osnabrück
Teutoburger Wald
Weserbergland
Münster-land
Münster
Bielefeld
Hildesheim
Salzgitter
Magdeburg
Fläming
Spree-wald
Brocken 1.141 m
Harz
Börde
Dessau-Roßlau
Lutherstadt Wittenberg
Cottbus
Niederlausitz
Niederrhein
Paderborn
Hamm
Bochum
Dortmund
Duisburg
Essen
Hagen
Wuppertal
Sauerland
Rothaargebirge
Langenberg 843 m
Göttingen
Thüringer Becken
Halle (Saale)
Leipziger Tieflandbucht
Leipzig
Kutschenberg 201 m
Oberlausitz
Dresden
Kassel
Mönchengladbach
Leverkusen
KÖLN
Düsseldorf
Hessisches Bergland
Eisenach
Gotha
Weimar
Erfurt
Jena
Gera
Chemnitz
Zwickau
Erzgebirge
Fichtelberg 1.244 m
Aachen
Bonn
Siegen
Marburg
Westerwald
Wetzlar
Gießen
Vogelsberg
Wasserkuppe 950 m
Rhön
Thüringer Wald
Gr. Beerberg 983 m
Hohe Acht 747 m
Neuwied
Koblenz
Eifel
Taunus
Frankfurt am Main
Wiesbaden
Offenbach a. Main
Mainz
Darmstadt
Spessart
Fichtelgebirge
Schneeberg 1.051 m
Bayreuth
Bamberg
Würzburg
Hunsrück
Erbeskopf 816 m
Trier
LUXEMBURG
Worms
Odenwald
Steigerwald
Dollberg 695 m
Ludwigshafen a. Rhein
Mannheim
Kaiserslautern
Heidelberg
Erlangen
Fürth
Nürnberg
Oberpfälzer Wald
Fränkische Alb
Saarbrücken
Heilbronn
Hesselberg 689 m
Großer Arber 1.456 m
Regensburg
Bayerischer Wald
Karlsruhe
Ludwigsburg
Pforzheim
Stuttgart
Ingolstadt
Oberrhein
Schwarzwald
Tübingen
Reutlingen
Schwäbische Alb
Landshut
Passau
Villingen-Schwenningen
Ulm
Augsburg
FRANKREICH
Freiburg im Breisgau
Feldberg
Lemberg 1.115 m
Alpenvorland
MÜNCHEN
Kempten (Allgäu)
Rosenheim
Konstanz
Bodensee
Schwarzer Grat 1.119 m
Allgäu
Hochfrottspitze 2.649 m
Zugspitze 2.962 m
Bayerische Alpen
Watzmann 2.713 m
ÖSTERREICH
LIECHTEN-STEIN
SCHWEIZ
ITALIEN
BELGIEN
0 25 50 75 100 km

# PERCENTAGE OF MOUNTAIN AREA PER COUNTRY




Criteria for mountains (*as defined by UNEP-WCMC, 2002*):

- Elevation > 300 m & local elevation range > 300 m
- Elevation > 1 000 m & slope > 5°
- Elevation > 1 500 m & slope > 2°
- Elevation > 2 500 m

Mountain area as percentage of total country area

- Less than 20%
- From 20% to 40%
- From 40% to 60%
- From 60% to 80%
- More than 80%

Source: Analysis by GRID-Arendal, 2016. Mountains derived from US Geological Survey National Mapping Division, EROS Data Center (EDC) (1996) GTOP030.

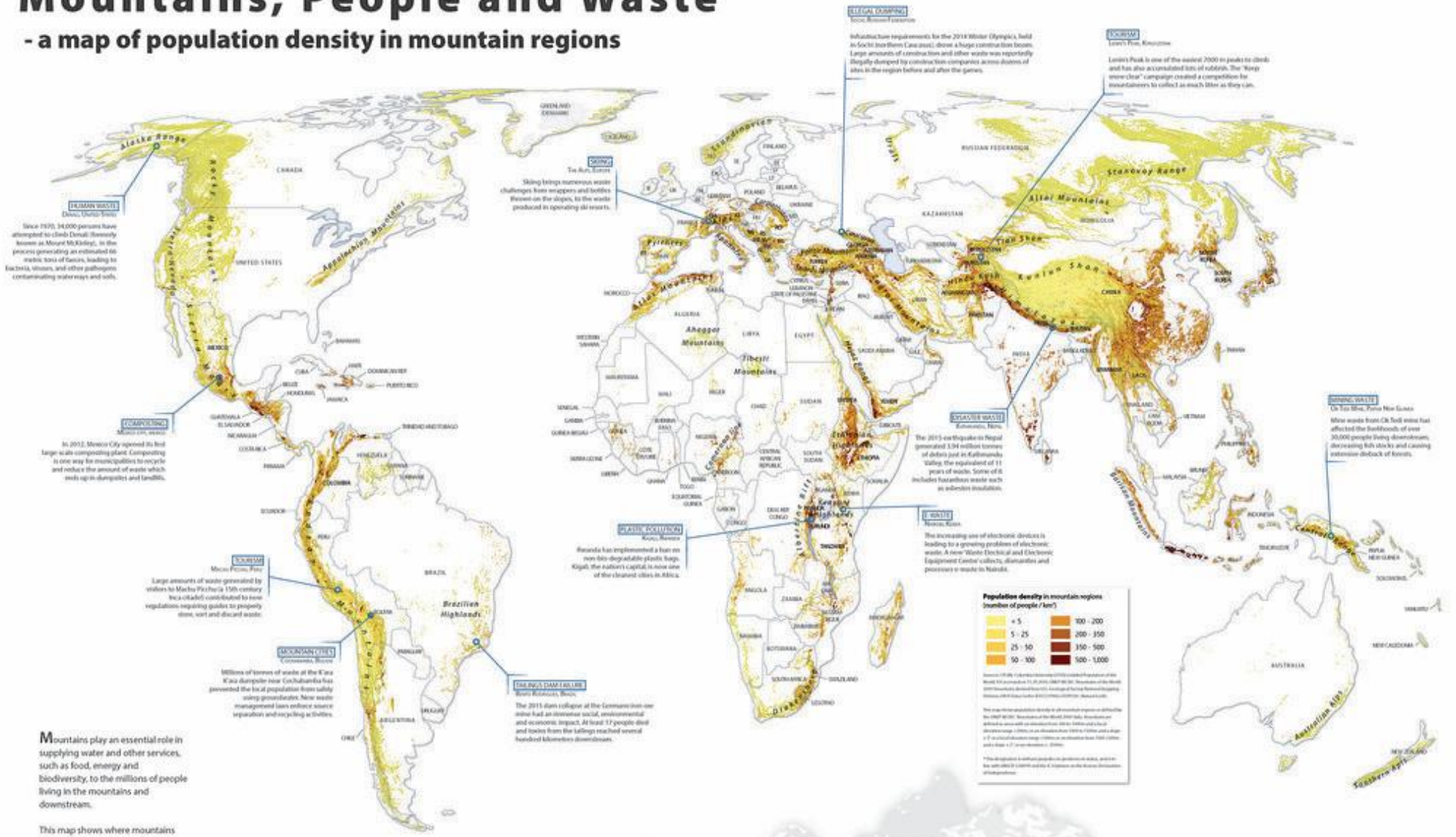
** This designation is without prejudice to positions on status, and is in line with UNSCR 1244/99 and the ICJ Opinion on the Kosovo Declaration of independence.*

*** Former Yugoslav Republic of Macedonia*

https://www.grida.no/resources/5032

# Mountains, People and Waste

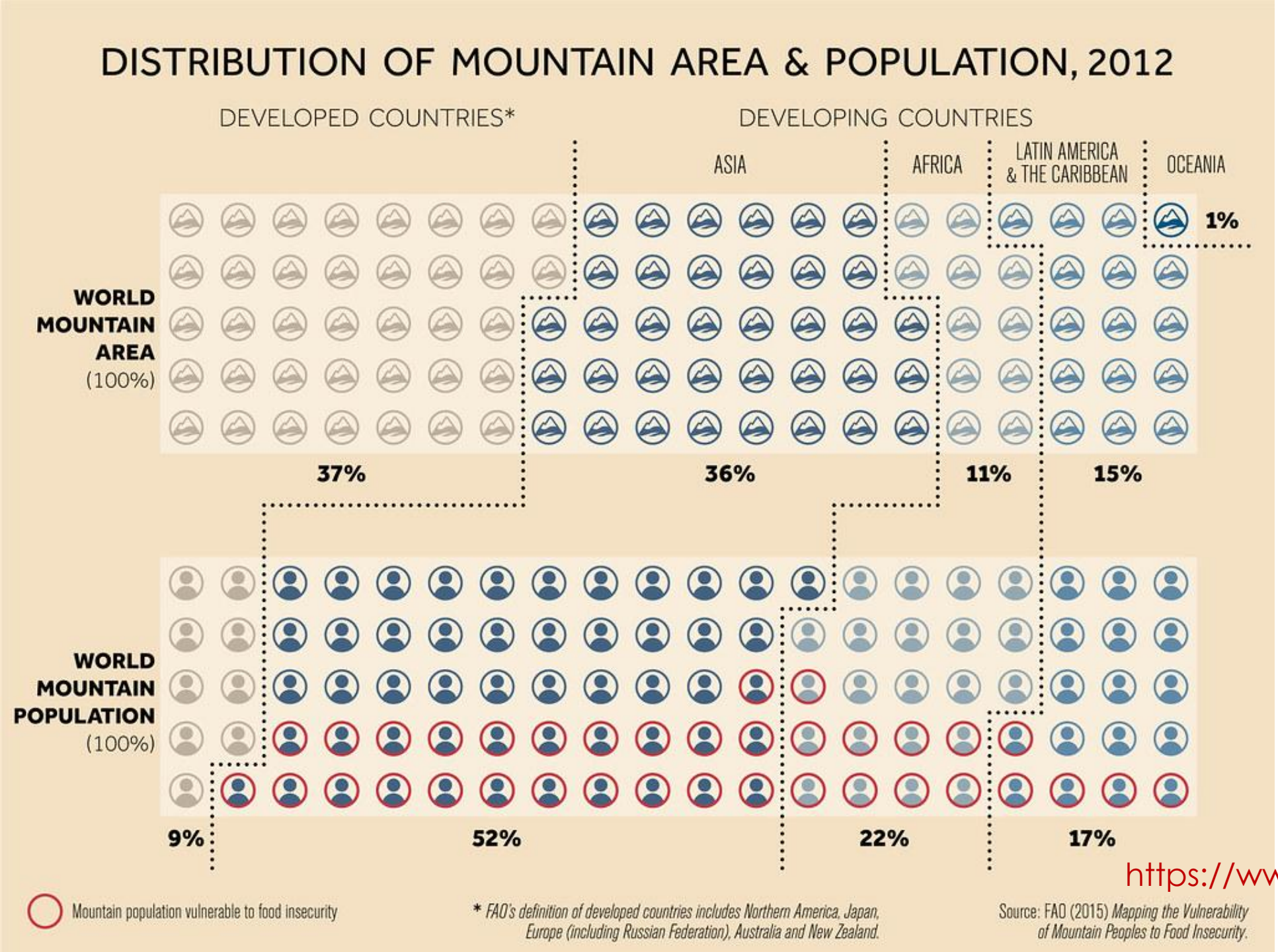
## - a map of population density in mountain regions




Mountains play an essential role in supplying water and other services, such as food, energy and biodiversity, to the millions of people living in the mountains and downstream.

This map shows where mountains are located around the world and the human population densities within these regions. Short case studies, which are referred to in the accompanying *Waste Management Outlook for Mountain Regions - Sources and Solutions*, highlight the variety of waste challenges and opportunities in mountain regions.

https://www.grida.no/resources/5040




https://www.grida.no/resources/533

# General definitions

Global estimates of **mountain populations**:

In 2015 estimates range between 0.344 billion (<5%) to 2.289 billion (>30%) (Thornton et al. (2022)).

Global estimates for people at risk of glacial lake outburst floods (GLOFs):

2023: 15 million (Taylor et al. 2023)

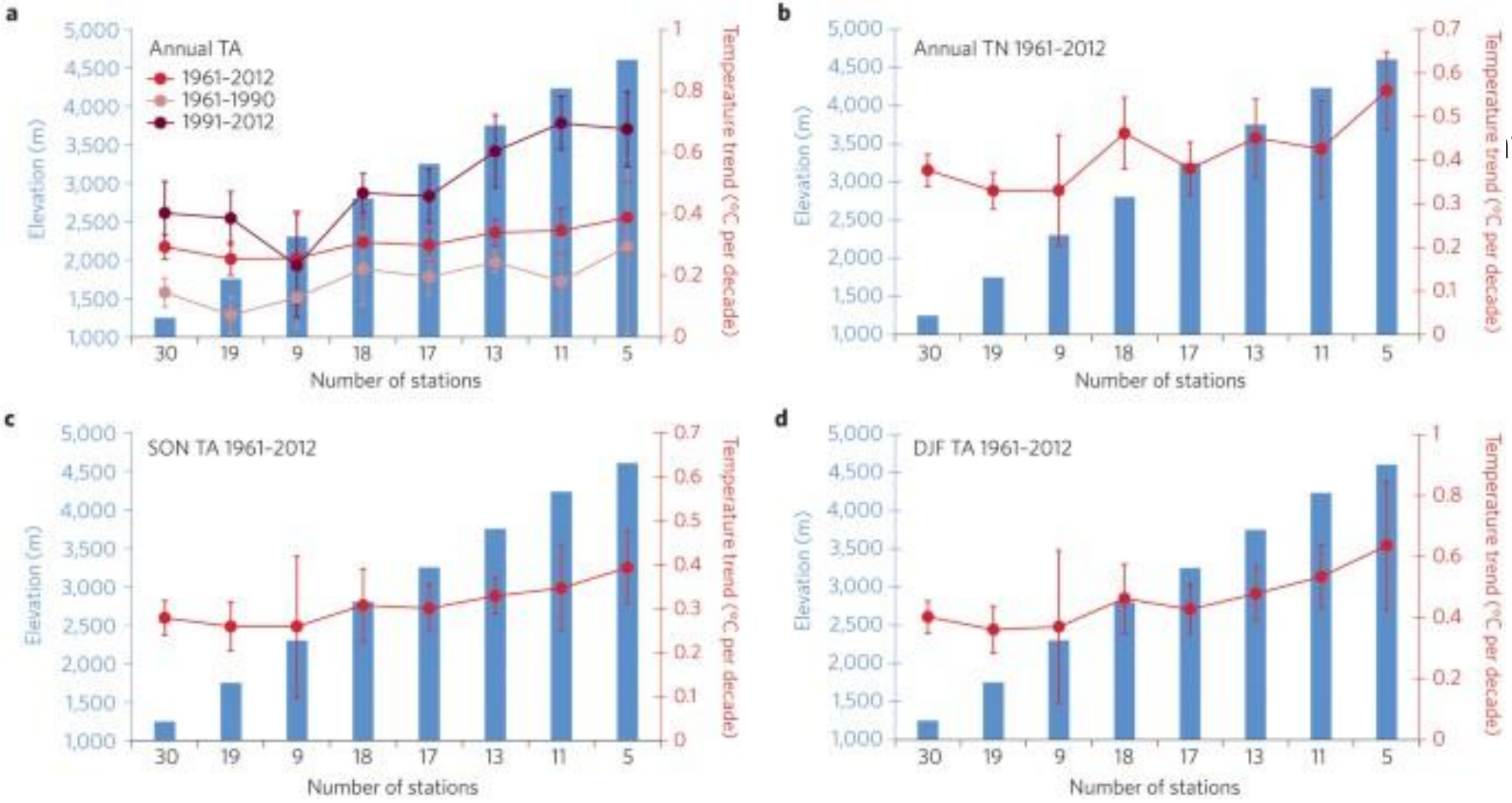
2024: >10 million (Zhang et al. 2024)

# why mountains and risk?

climate:

- climate change happens faster at altitude (concept of ‘elevation dependent warming’ EDW now shifting to ‘elevation dependent climate change’ EDCC)

# why mountains and risk?




t warming'

Pepin et al. (2015)

# why mountains and risk?

climate:

- climate change happens faster at altitude (concept of ‘elevation dependent warming’ EDW now shifting to ‘elevation dependent climate change’ EDCC)
- climate variability is much stronger with rough topography

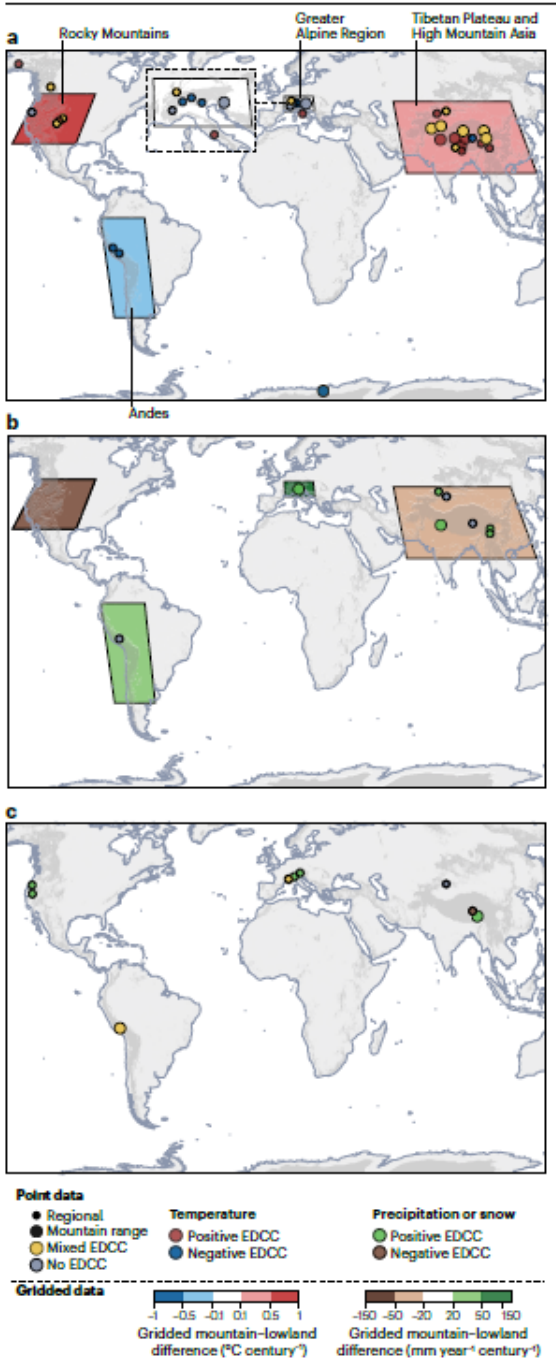



**Fig. 5 | Synthesis of observed elevation-dependent climate change.** **a**, Published evidence of elevation-dependent warming for individual mountain studies (small circles, representing only the sign; Supplementary Table 8) and mountain range-scale studies (larger circles, representing only the sign; the location of the circle is approximate; Supplementary Table 8), and the mean warming signal inferred from gridded datasets for entire mountain regions (large boxes, the size indicating the spatial scale for averaging, colour representing the sign and magnitude of the change; Table 1 and Supplementary Tables 1–3). The insert provides a more detailed view of the European Alps. **b**, As in panel **a**, but for precipitation. **c**, As in panel **a**, but for snow and without the gridded datasets. Patterns of elevation-dependent climate change (EDCC) are mixed and depend on the variable and region considered.

EDPC trends differ between mountain ranges and even between individual studies in the same range. There is a lack of agreement in EDPC trends from individual studies and the gridded datasets over HMA and the TP, with observational studies showing mostly positive EDPC contrasting with mixed results from the gridded datasets (Fig. 5b). In the GAR, positive EDPC dominates in both gridded data and individual studies. Observational studies using variables such as snow depth, snowfall and snow cover find that snow is almost universally decreasing, with most snow loss occurring at lower mountain elevations (Fig. 5c). Often in studies focused in high mountains, the most rapid snow loss occurs only up to a certain elevation, above which snow is typically more stable. This finding seems to contradict the decreasing orographic snowfall gradient identified in nearly all regions in the gridded analyses, which ranges from −5 mm $yr^{-1}$ $century^{-1}$ (CRU, TP and HMA) to −91 mm $yr^{-1}$ $century^{-1}$ (ERA5, GAR) (Table 1). However, the gridded analysis is solely based on snowfall, and it takes mountains as a whole and compares the mountain snowfall loss with lowland regions where snowfall is usually a scarcity. Thus, more loss is observed in the higher-elevation band, reducing the orographic gradient.

## Summary and future perspectives

Contrasting warming rates at differing elevations are usually associated with changes in elevation gradients of many climate variables, including albedo, precipitation, atmospheric moisture, cloud, radiation balance components and wind speed; therefore, the concept of EDW has been expanded to EDCC. Systematic changes in these gradients are commonly associated with land cover changes, including uphill shifts of snowlines and treelines, the intensification of the hydrological cycle, and changes from snow to rain. Observed EDW can often be attributed to drivers such as surface effects (including SAF) and free-air effects (atmospheric humidity and DLR) that depend on season and time of day. There are regional differences in observed EDCC over the historical period (1950 to present), including a latitudinal contrast, contrasts within mountain ranges (leeward versus windward slopes), and differences due to mountain orientation and topography. Climate models simulate future EDCC throughout the twenty-first century that is characterized by strong regional variation, due to the variable impact of different drivers in diverse regions and over different timescales. Further work is needed to better understand the trends, drivers and implications of EDCC, as discussed later in this section.

### Limited coverage of mountain in situ observations

Current in situ observations are inadequate for understanding the complexities of EDCC. Observations are highly skewed towards lower elevations, and particularly biased towards mid-latitudes and the Northern

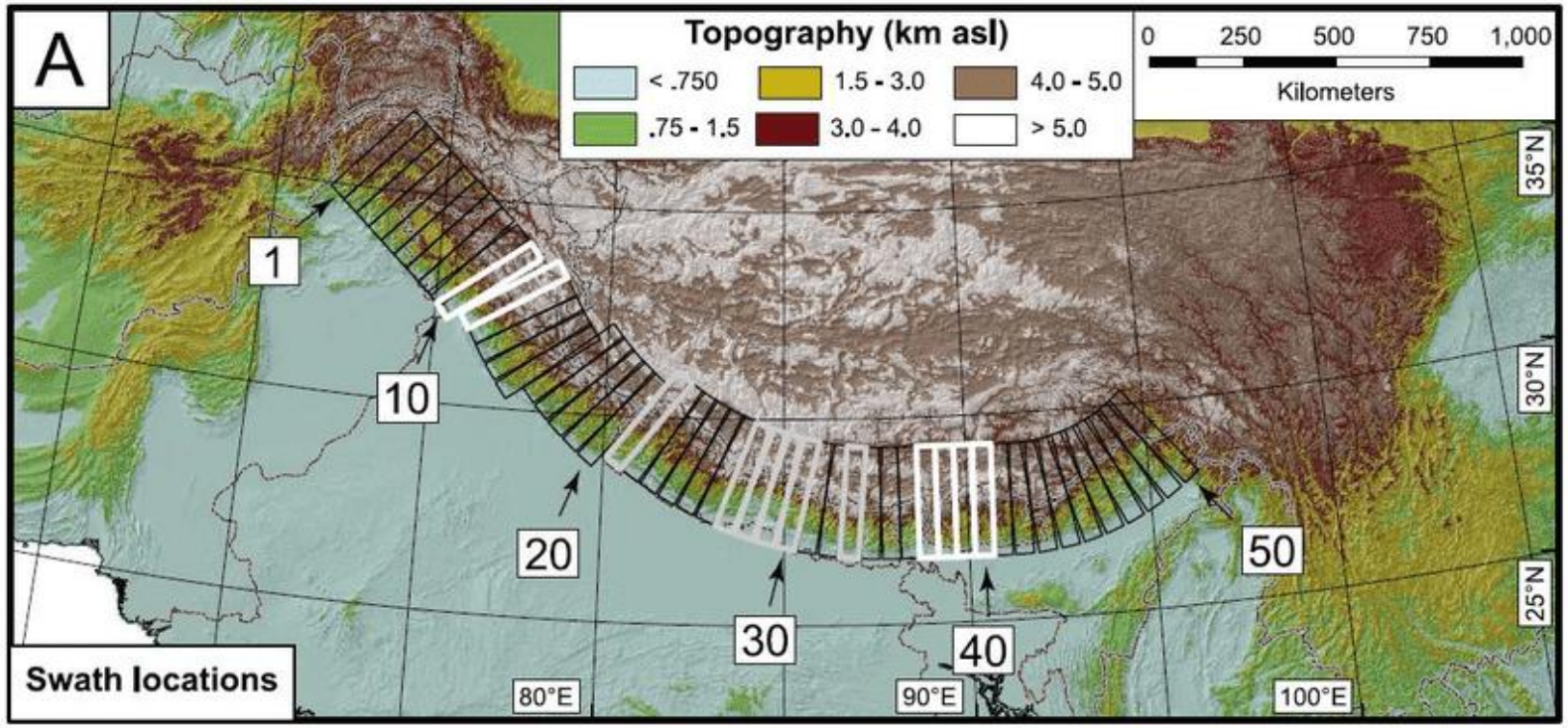


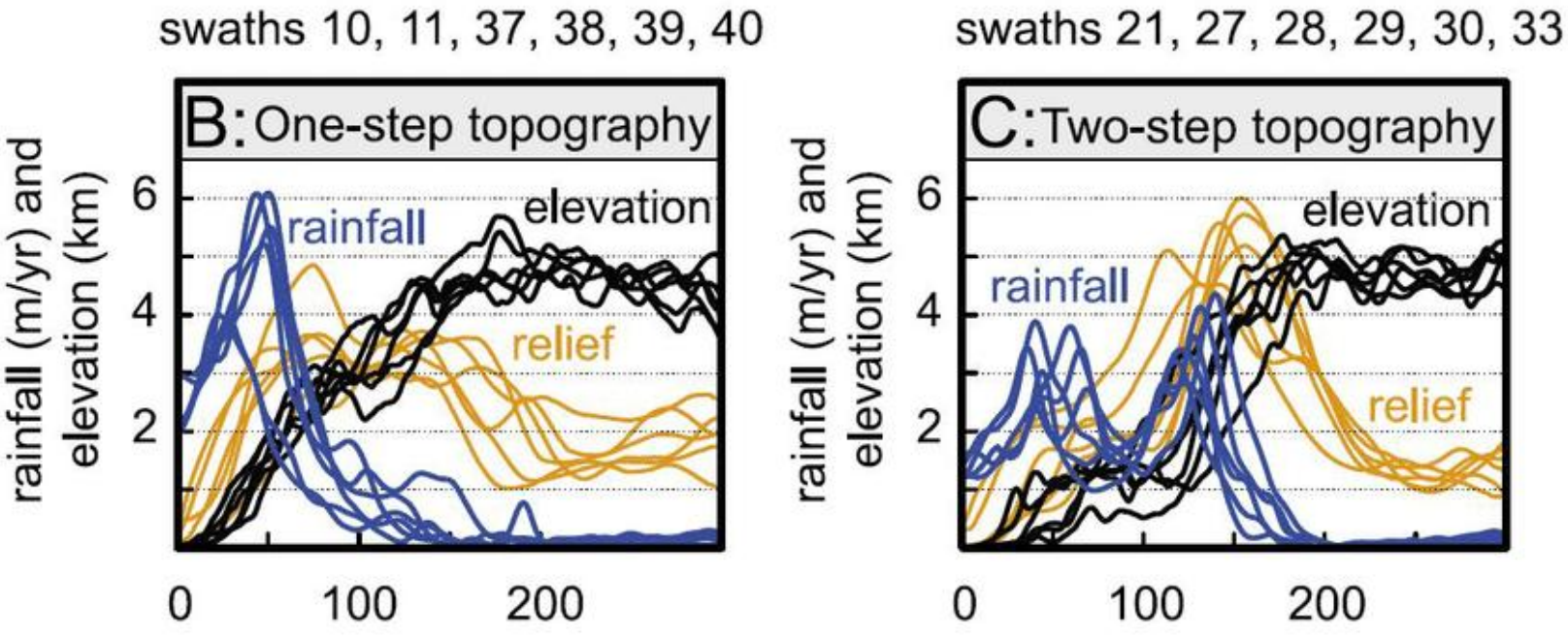
Pepin et al. (2025)

# why m

climate:

- climate ch … t warming'
  EDW now …
- climate vc …




Veh (2019)

# wh

clim

- cl ng’
  ED
- cl



# why mountains and risk?

climate:

- climate change happens faster at altitude (concept of ‘elevation dependent warming’ EDW now shifting to ‘elevation dependent climate change’ EDCC)
- Climate variability is much stronger with rough topography

politics/governance:

- Mountain regions tend to be more ‘remote’ in terms of governance and access
- Mountain regions are often located along national borders, creating challenges of (transboundary) management

→ Heightened vulnerability comparable to small island states

# General definitions

***risks*:**

How do you define it?

# General definitions

***risks*:**

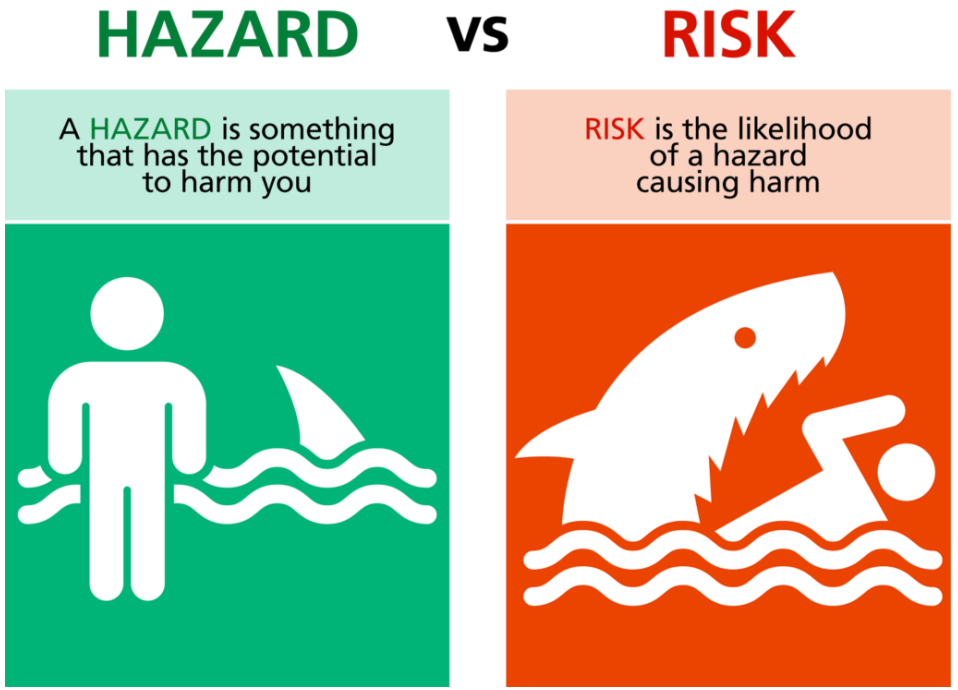
*A measure of the likelihood or probability that damage to life, health, property, and/or* the environment will occur as a result of a particular hazard.

(Exposure to) the possibility of loss, injury, or other adverse or unwelcome circumstance; a chance or situation involving such a possibility

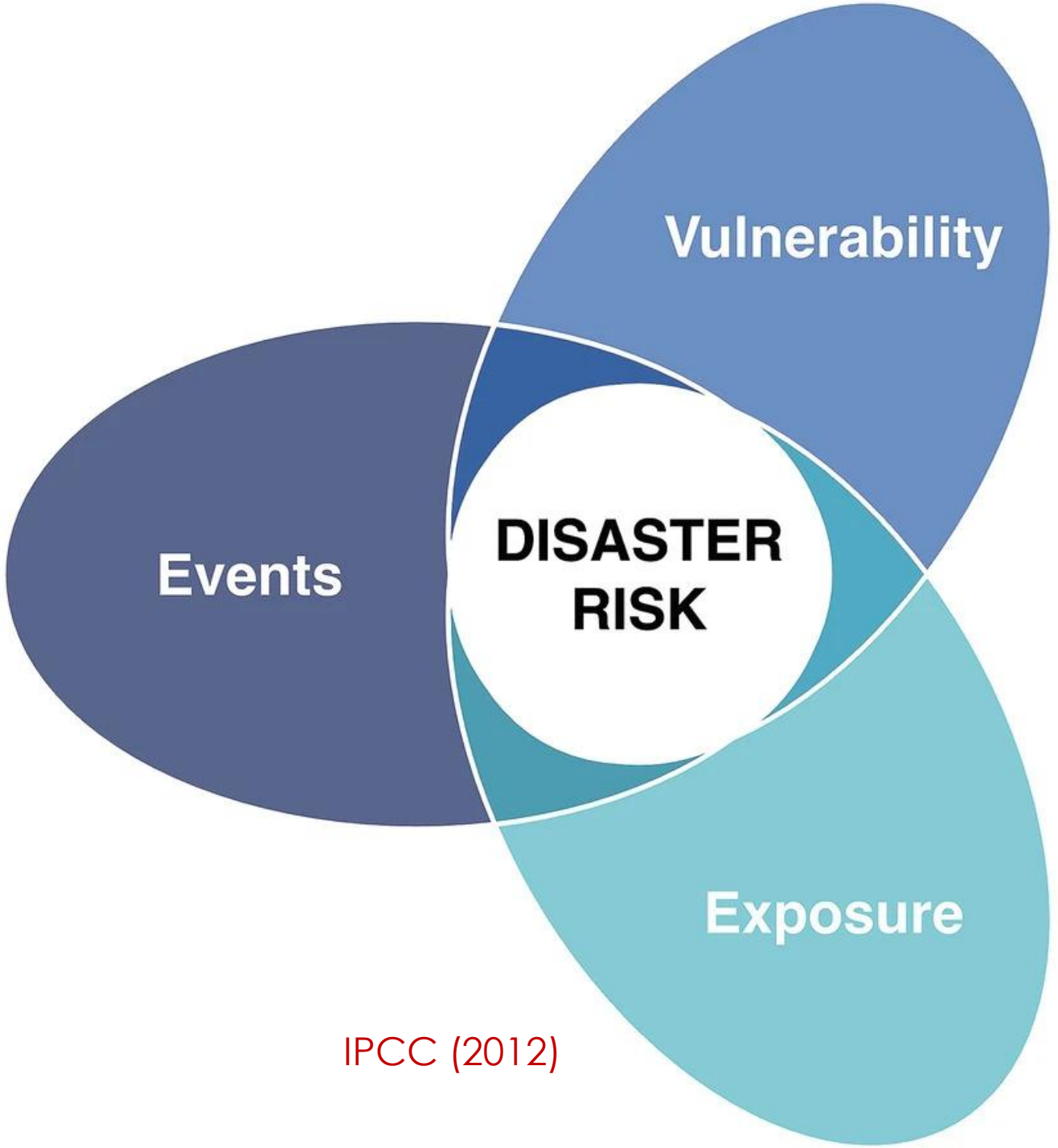
(Oxford Dictionary)

*It can mean both the possibility of danger and*

*simultaneously its potential consequences.*

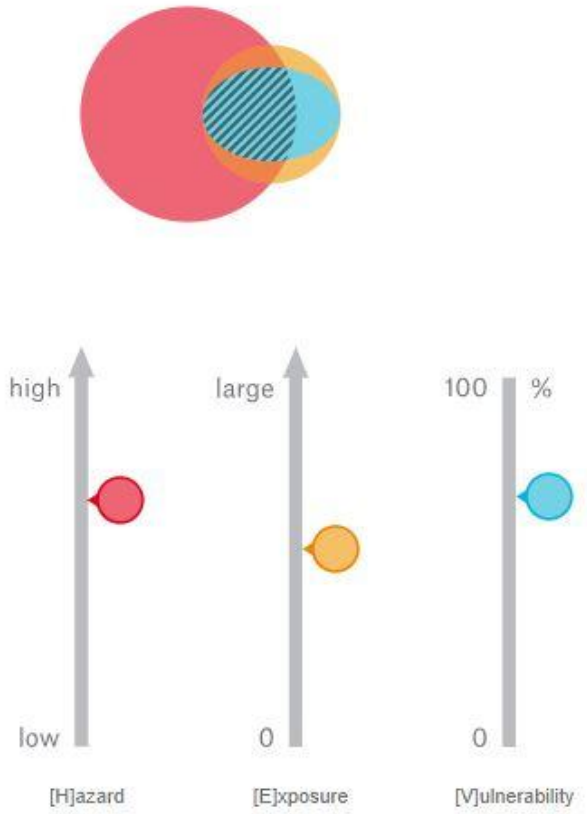


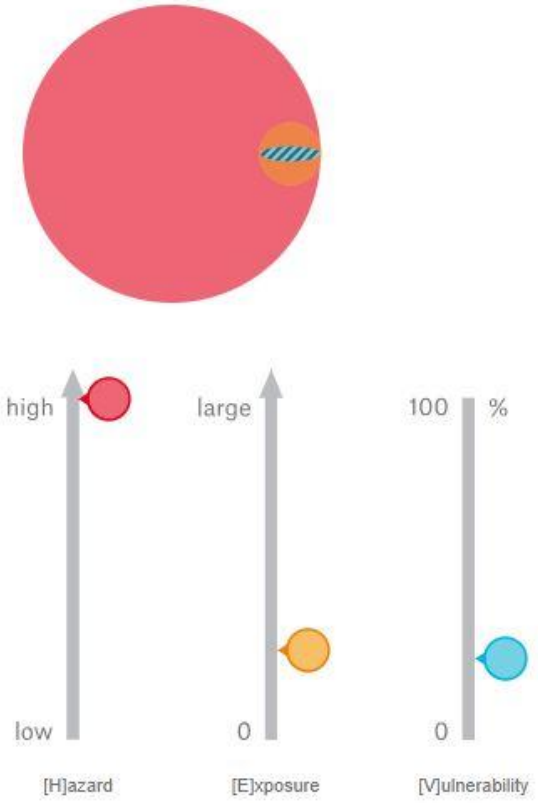
# General definitions




IPCC (2012)

# *risk = hazard * exposure * vulnerability [*adaptive capacity *insurance penetration]*




MunichRe

# General definitions

**Accepted risks**

"Some of the stakeholders are mainly impacted by **how frequently** they are affected by a damage or loss, **no matter how small** it might be. Other stakeholders are more affected by the **consequences of possible maximum impacts**. Some stakeholders might be more interested in the **probability of damages** reaching or **exceeding a certain threshold** or the damage at certain probabilities.

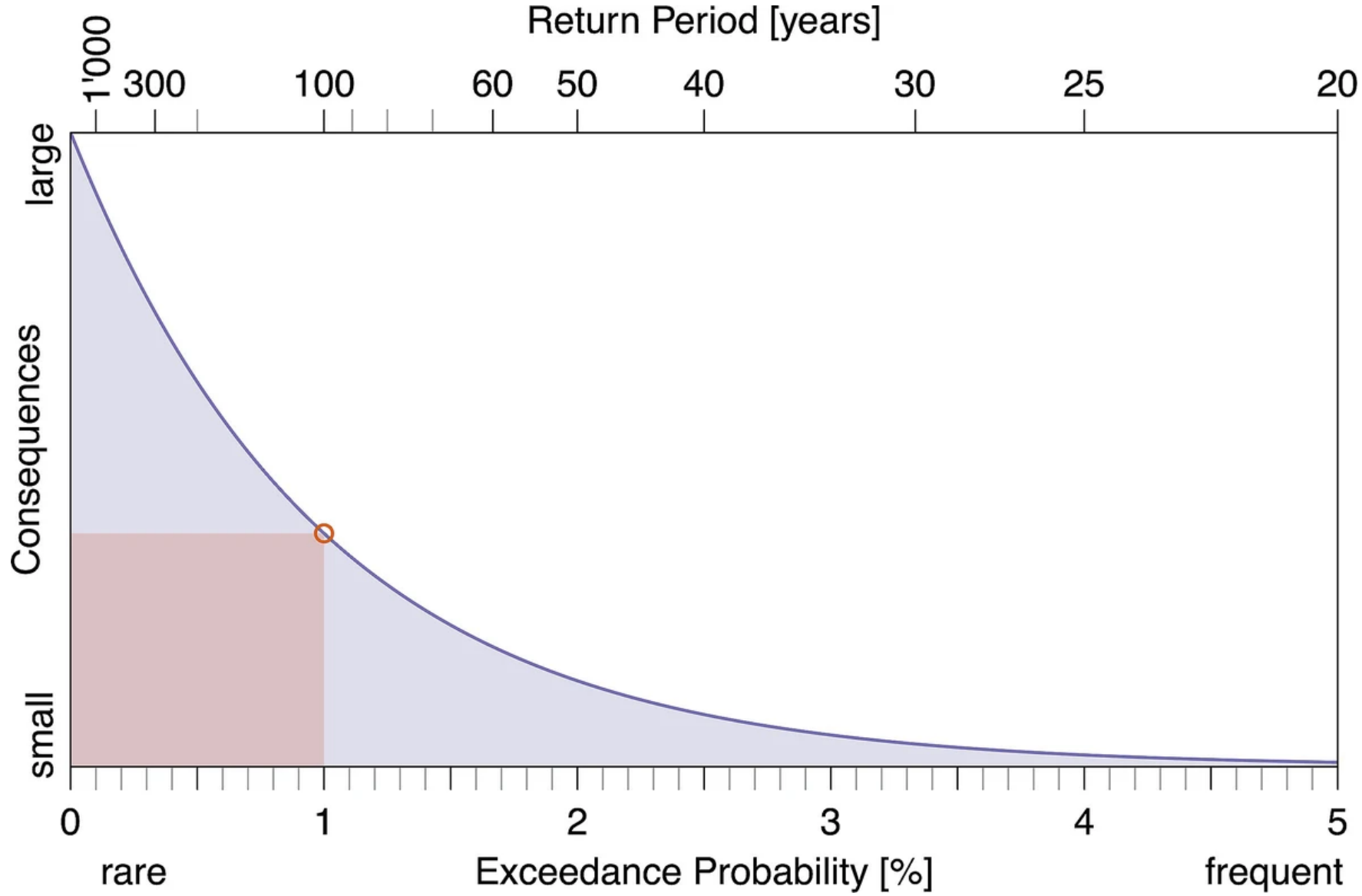
Furthermore, stakeholders might be interested in **different time horizons for their risk**. Some are interested in the risk within the next months to years, some rather within the next decades."

"For the operator of a street, the acceptable risk depends on the frequency of the street being flooded and the consequences of the flooding. It might be acceptable that the street is flooded once every other year as long as the street only has to be cleaned. For the operation of a regional medical emergency service, the level of accepted risk for the access road being flooded can depend on the availability of a redundant emergency service in the vicinity. If there is another emergency service in the vicinity, it is acceptable that the access road is being flooded as long as the interruption of the access is of short duration. If the emergency service is the only one in a larger region, a flooded access road is not acceptable." Kleinn et al. (2022)



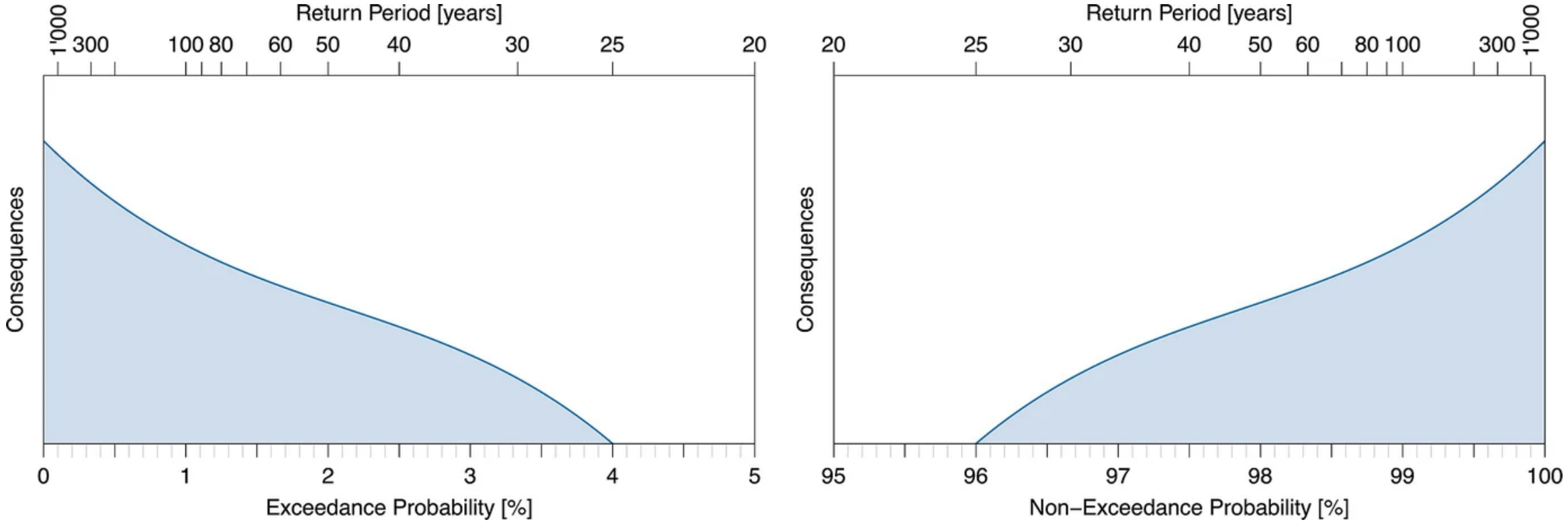
Community Interaction,
Melamchi, Nepal
May 2022

# Exceedance probability / loss (frequency) curve




Kleinn et al. (2022)

# Exceedance probability / loss (frequency) curve




Loss curves need many observations!

Kleinn et al. (2022)

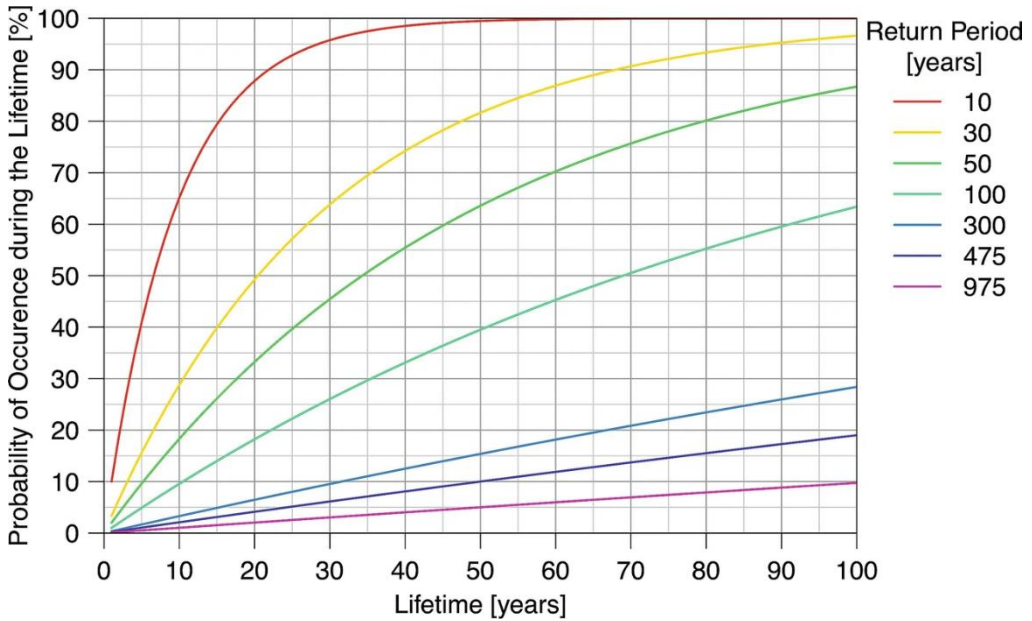
# Return period

The volume of this event at this location had a return period of 1 in 100 years ("100-year event")

$$Return\ Period = \frac{1}{Exceedance\ Probability} = \frac{1}{1 - Non\text{-}Exceedance\ Prob}$$

Probability for a certain time period given a return period

$$P = 1 - \left(1 - \frac{1}{RetPer}\right)^{Nyrs}$$




… what to do when return periods keep changing so fast?

# Characteristics of Risk

It is possible to characterize risk in numerical terms based on loss curves

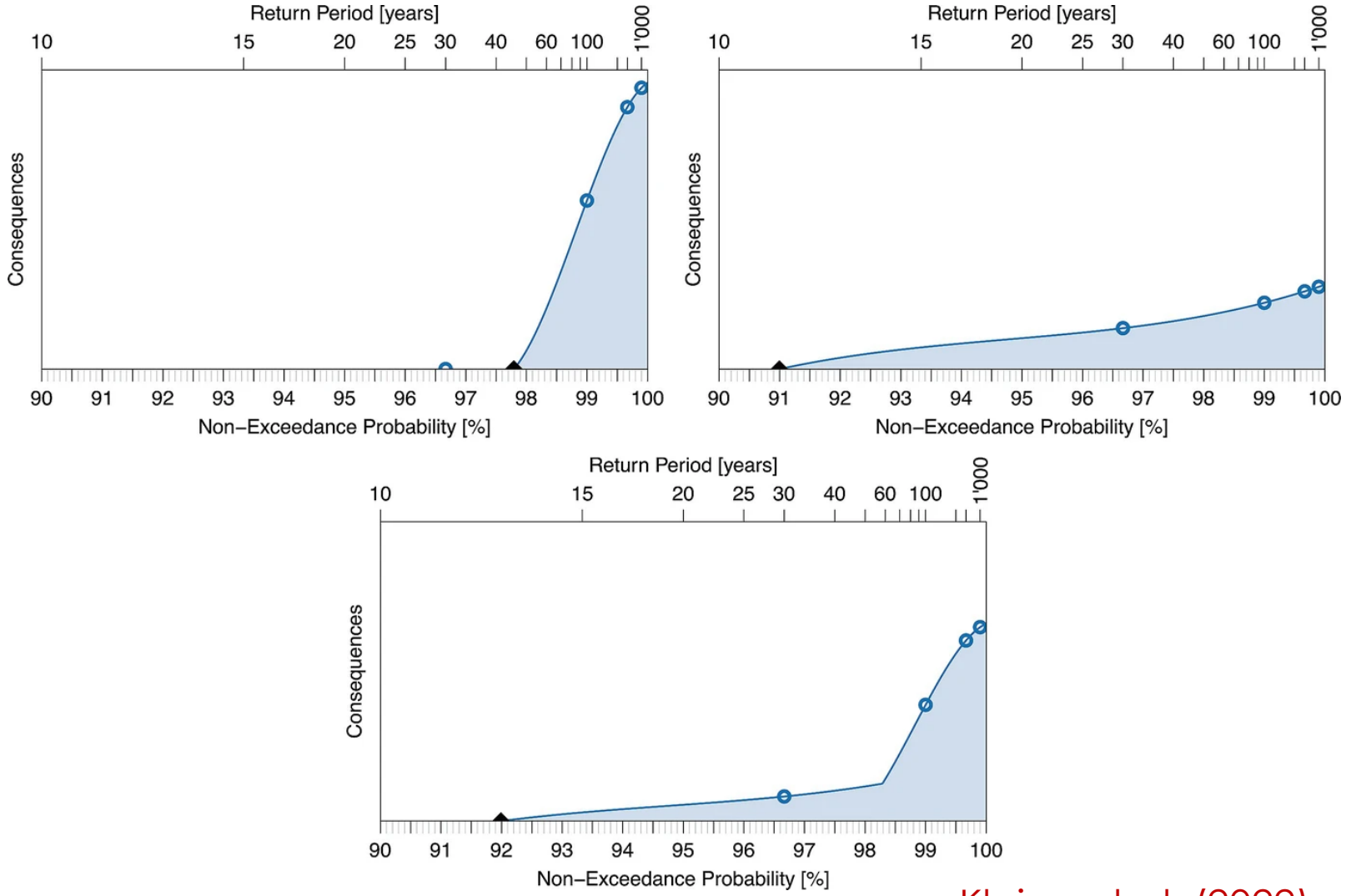
- The shape of the loss curve
- The probability of first loss
- The integral under the curve – the expected annual loss
- Specific loss values at return periods

# Characteristics of Risk

Top left: metropolitan area with rare natural hazard events which cause large losses

Top right: region with very low population density experiencing regular natural hazard events and a very limited maximum loss

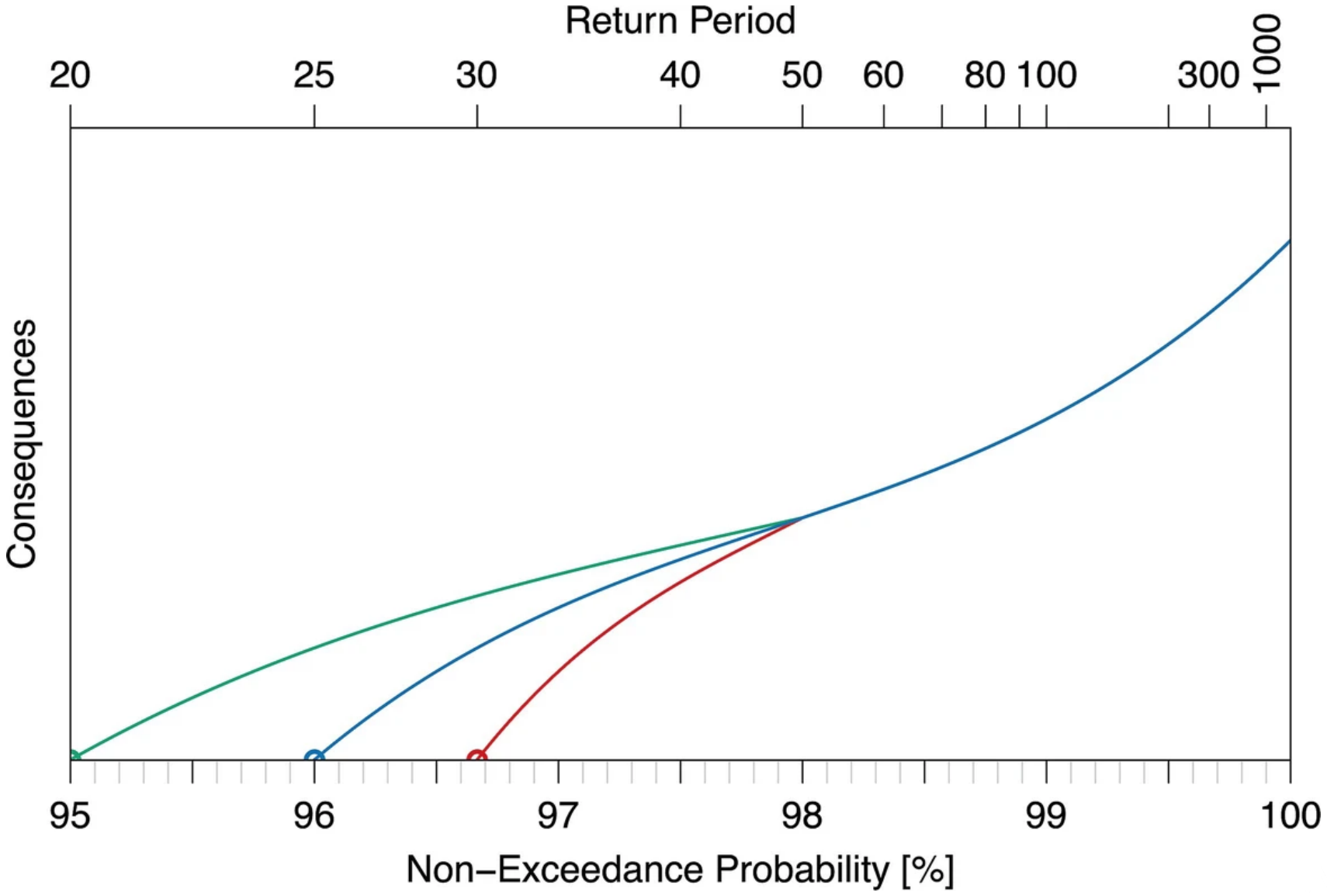
Bottom: a certain degree of protection is provided up to a certain intensity and therefore probability and losses increase rapidly thereafter




Kleinn et al. (2022)

# Occurrence of first loss

Knowledge about the occurrence of first loss has large impacts on the total expected annual loss (return period only changes slightly, but area under the integral changes a lot)

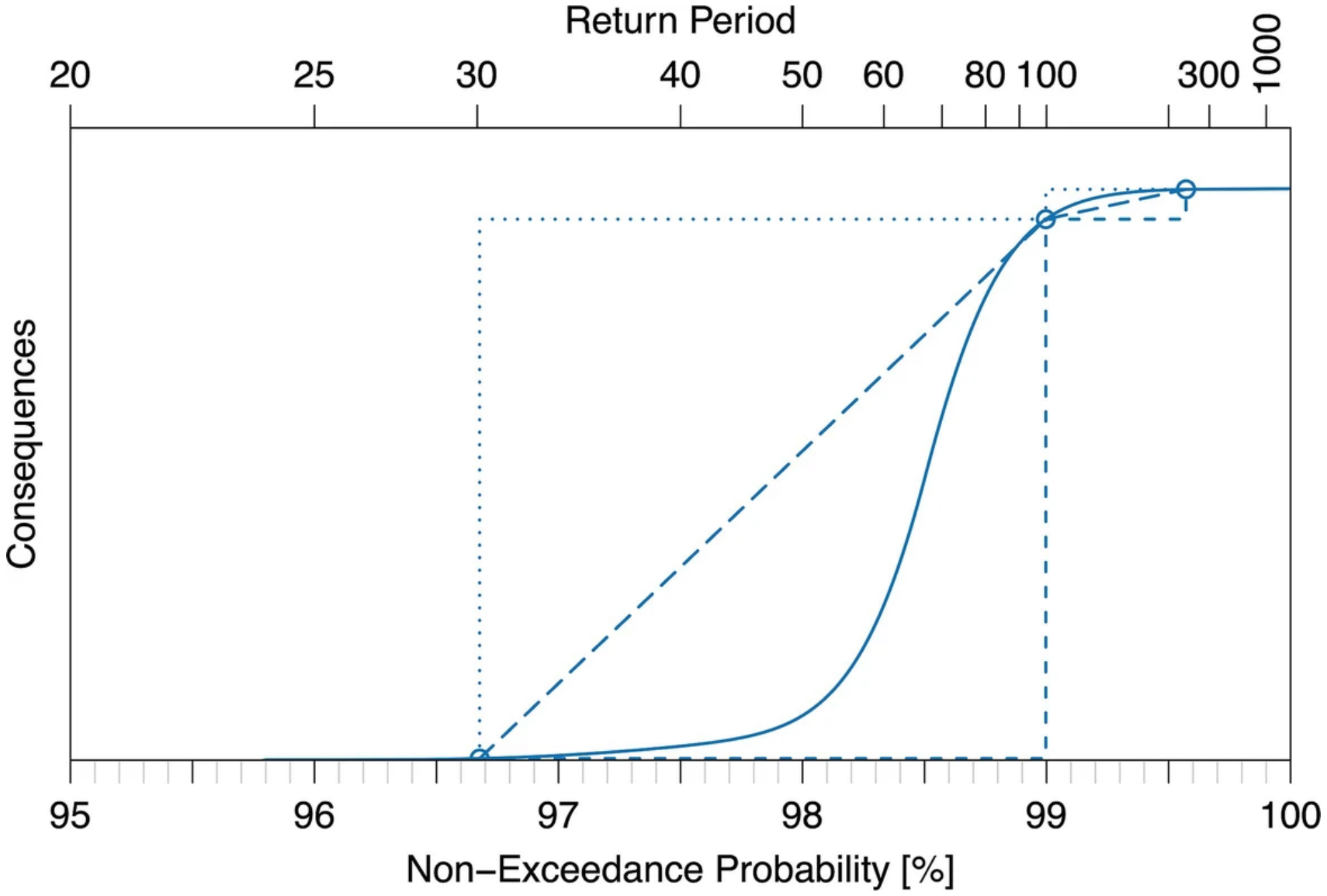



Kleinn et al. (2022)

# Uncertainty with limited data

If only few data points are available (dots) the total expected loss can not be precisely determined and can be very high (dotted) or very low (dashed line).

This will be a typical case for mountains or other vulnerable regions.




Kleinn et al. (2022)

# Uncertainty with limited data

*Hazards*: limited time series

*Exposure*: very limited comprehensive or accessible datasets (population, infrastructure)

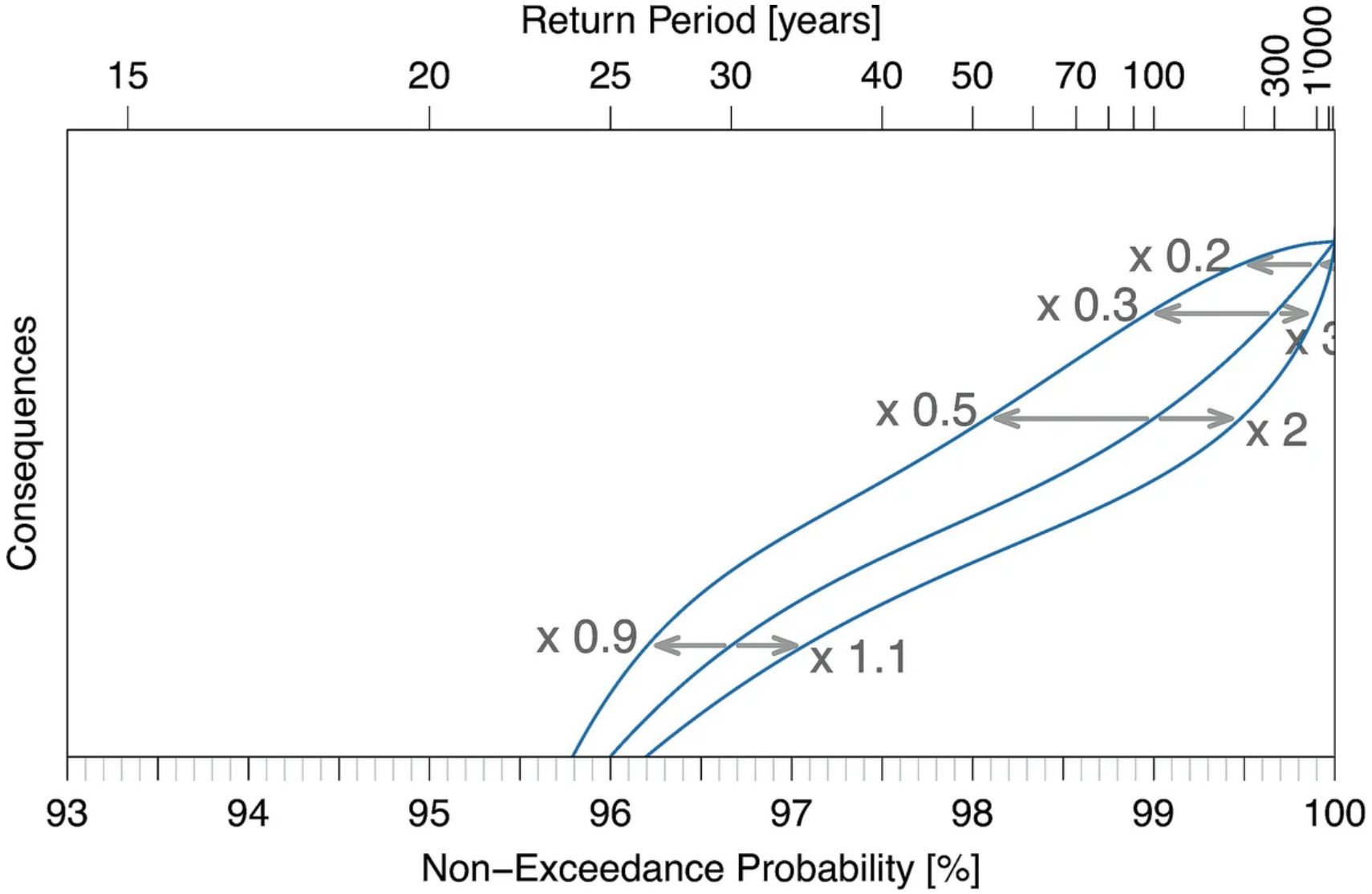
*Vulnerability*: Hard to pinpoint across regions and time

Kleinn et al. (2022)

# Uncertainty with limited data

Uncertainties generally larger with rare events

Large uncertainties for rare events have a smaller overall impact than small uncertainties for frequent events




Kleinn et al. (2022)

# General definitions

**Exposure**:

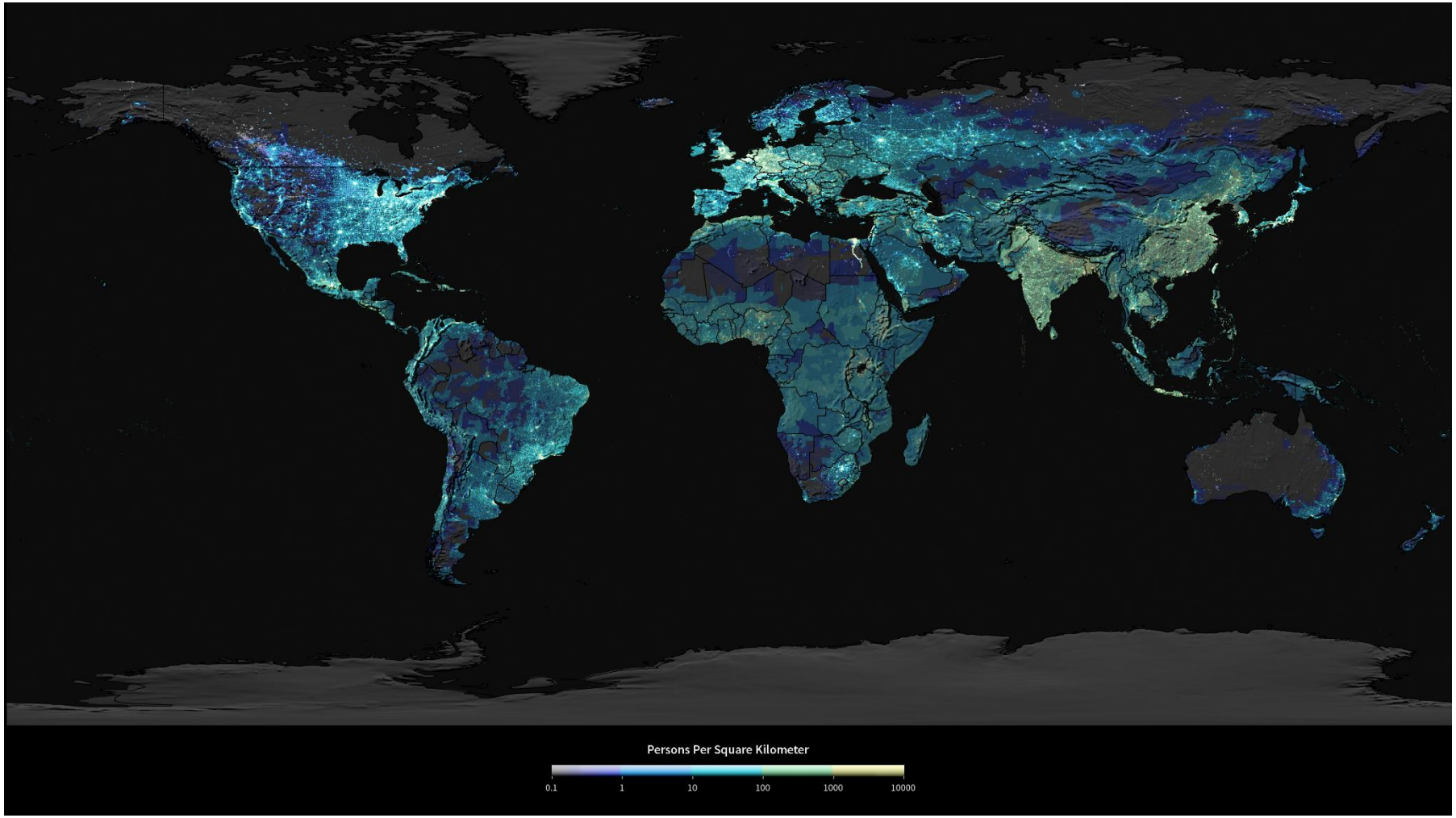
- *known assets – can be directly relevant to humans (e.g. houses, roads) or not (e.g. ecosystem)*
- *becomes more difficult with non-tangible assets (e.g. heritage, religious sites)*

- *assessments often done locally, but remote sensing has enabled big progress in large scale exposure data*
- *in mountains, remote sensing data is often not adequate and/or validation data is lacking*

# Global exposure

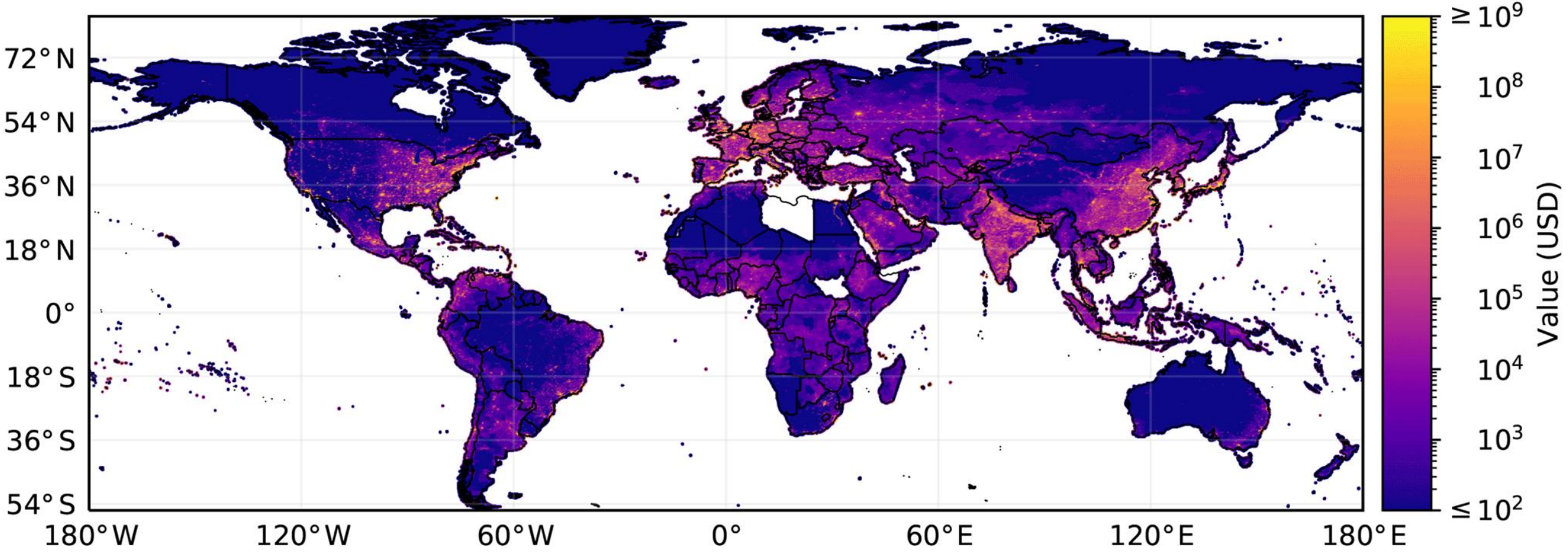
Using nightlights to estimate population densities

What could be drawbacks?



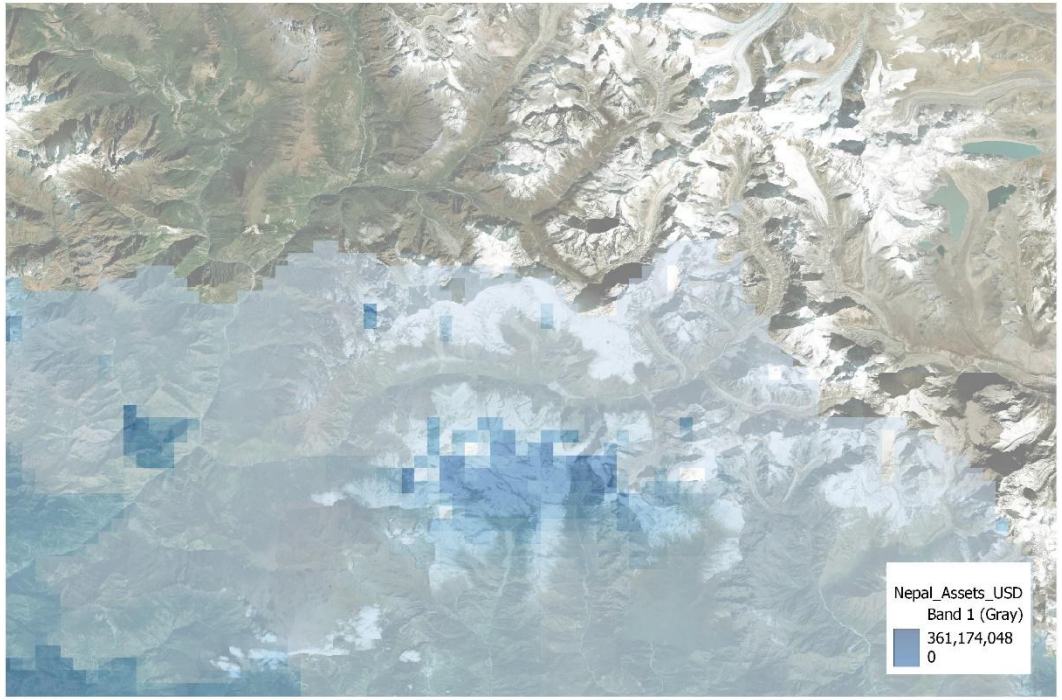

https://svs.gsfc.nasa.gov/30214

# Global exposure



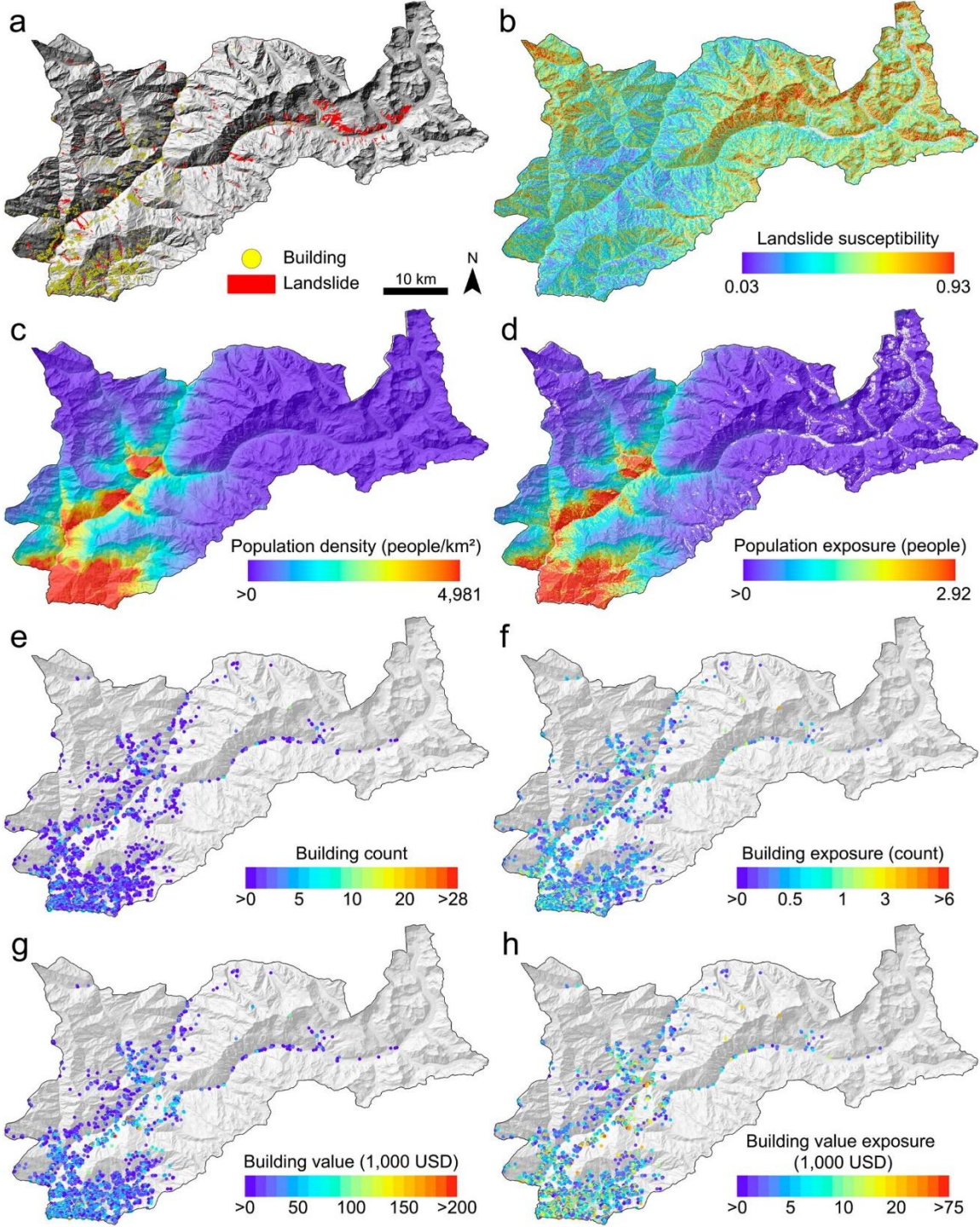

Eberenz et al. (2020)
CLIMADA global data
Nightlight intensity

# Local exposure




Eberenz et al. (2020)
CLIMADA global data
Nightlight intensity

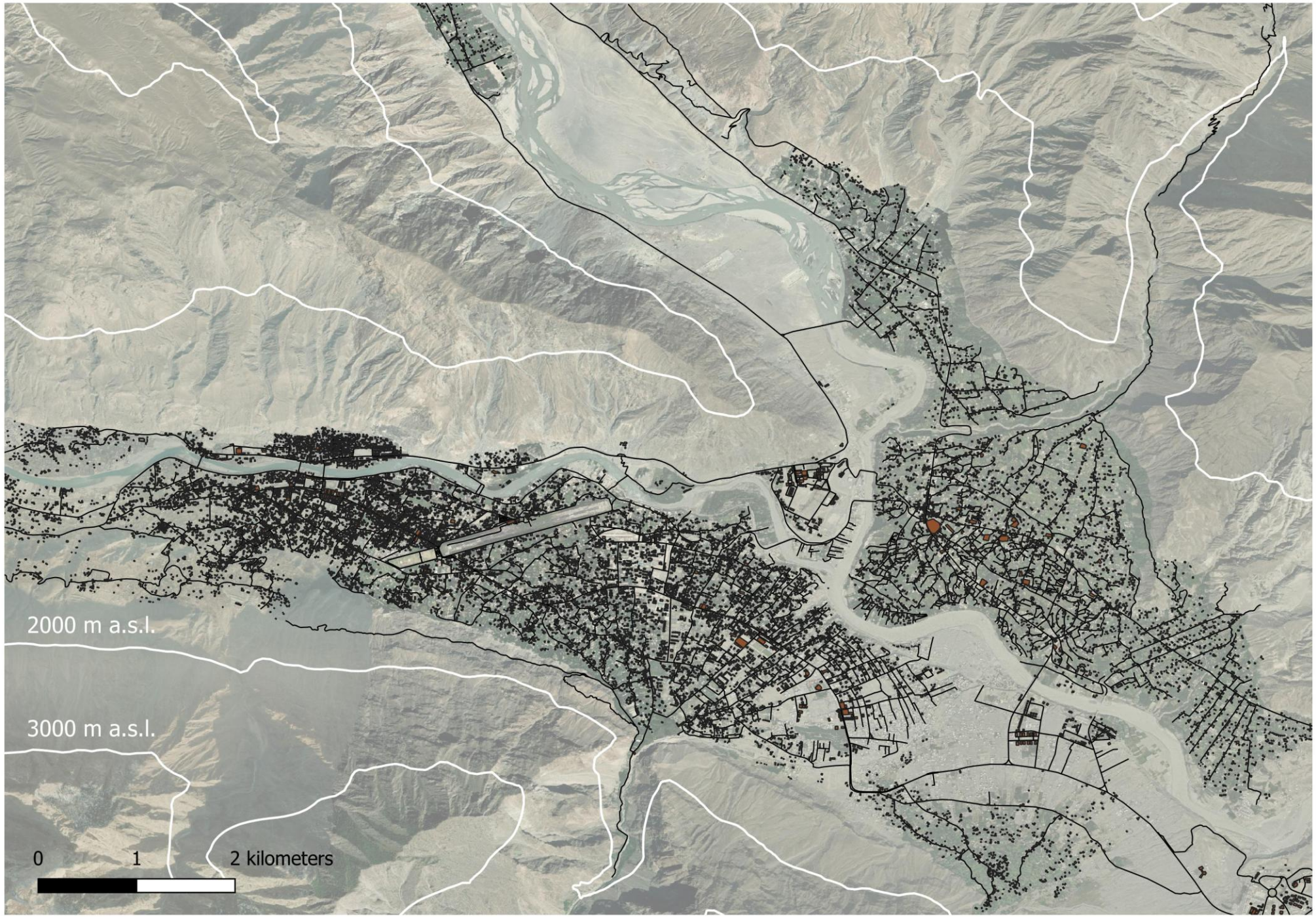
Kincey et al. (2024)
Meteor regional data
Optical/ML



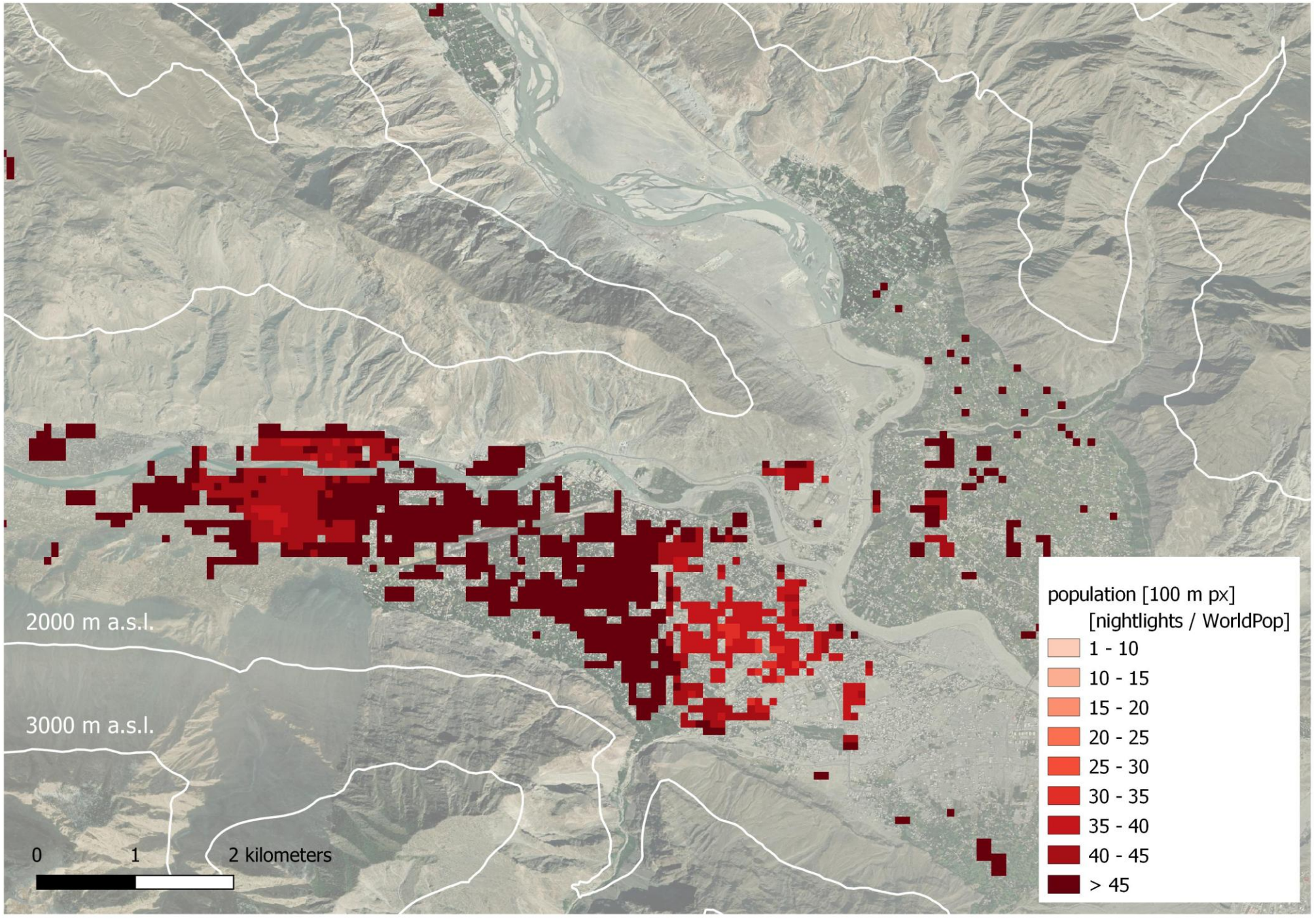



Gilgit
'Provincial' Capital
Gilgit-Baltistan
Pakistan

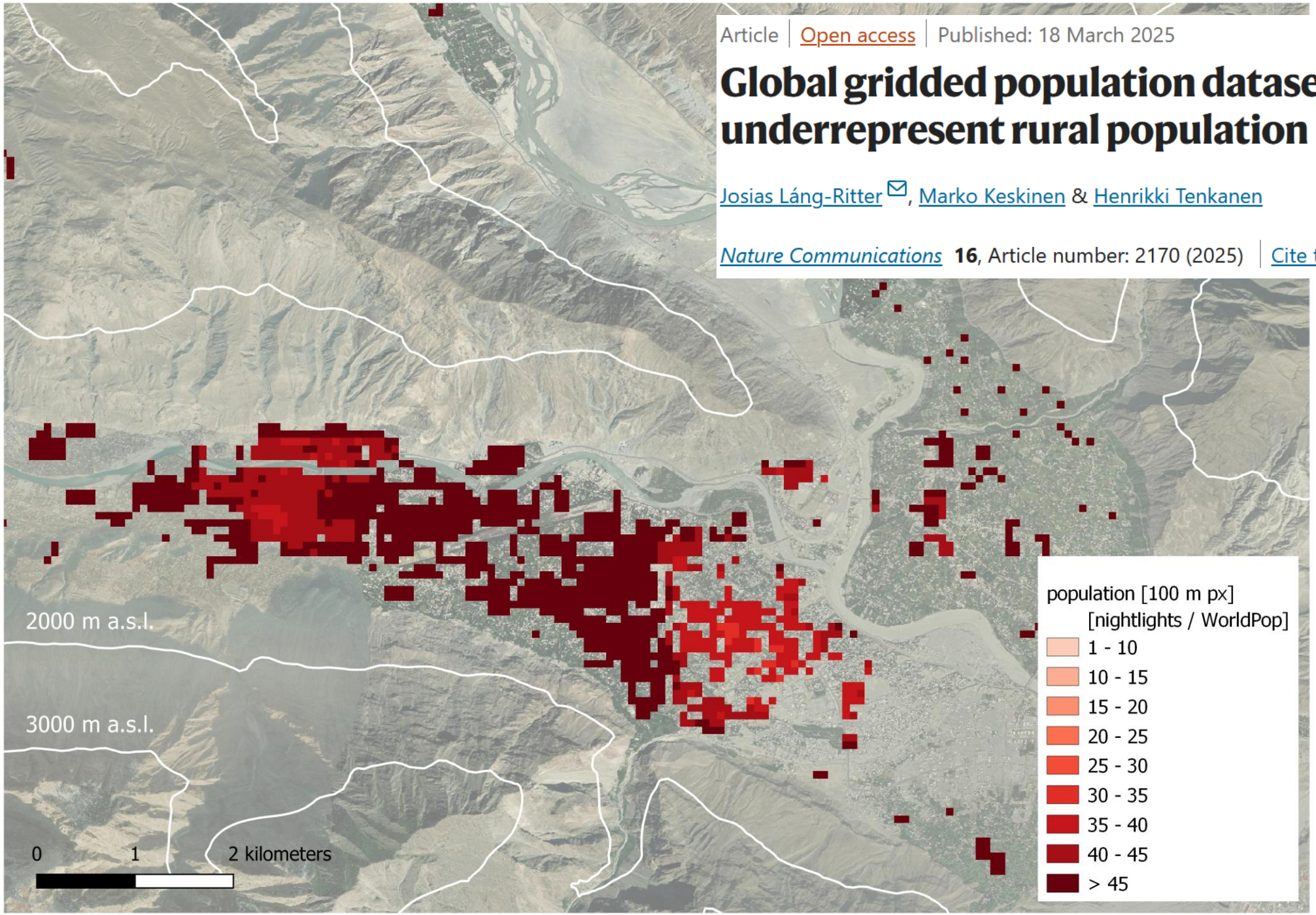
~315 000 inhabitants
~1500 - 1800 m a.s.l.




HOT-Open Street Ma
Data




WorldPop Population
data using nightlights
(2020)



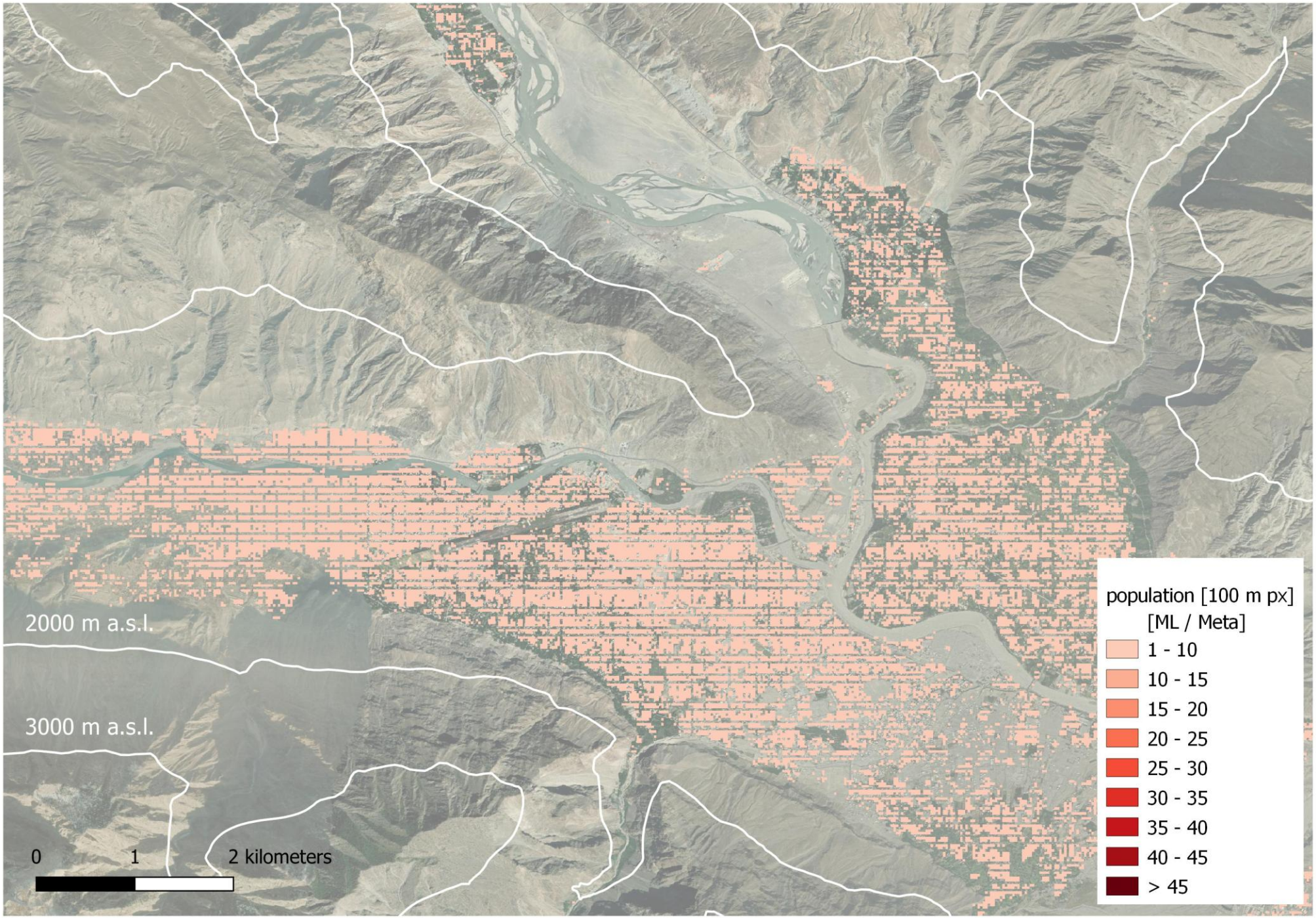

Article | Open access | Published: 18 March 2025

# Global gridded population datasets systematically underrepresent rural population

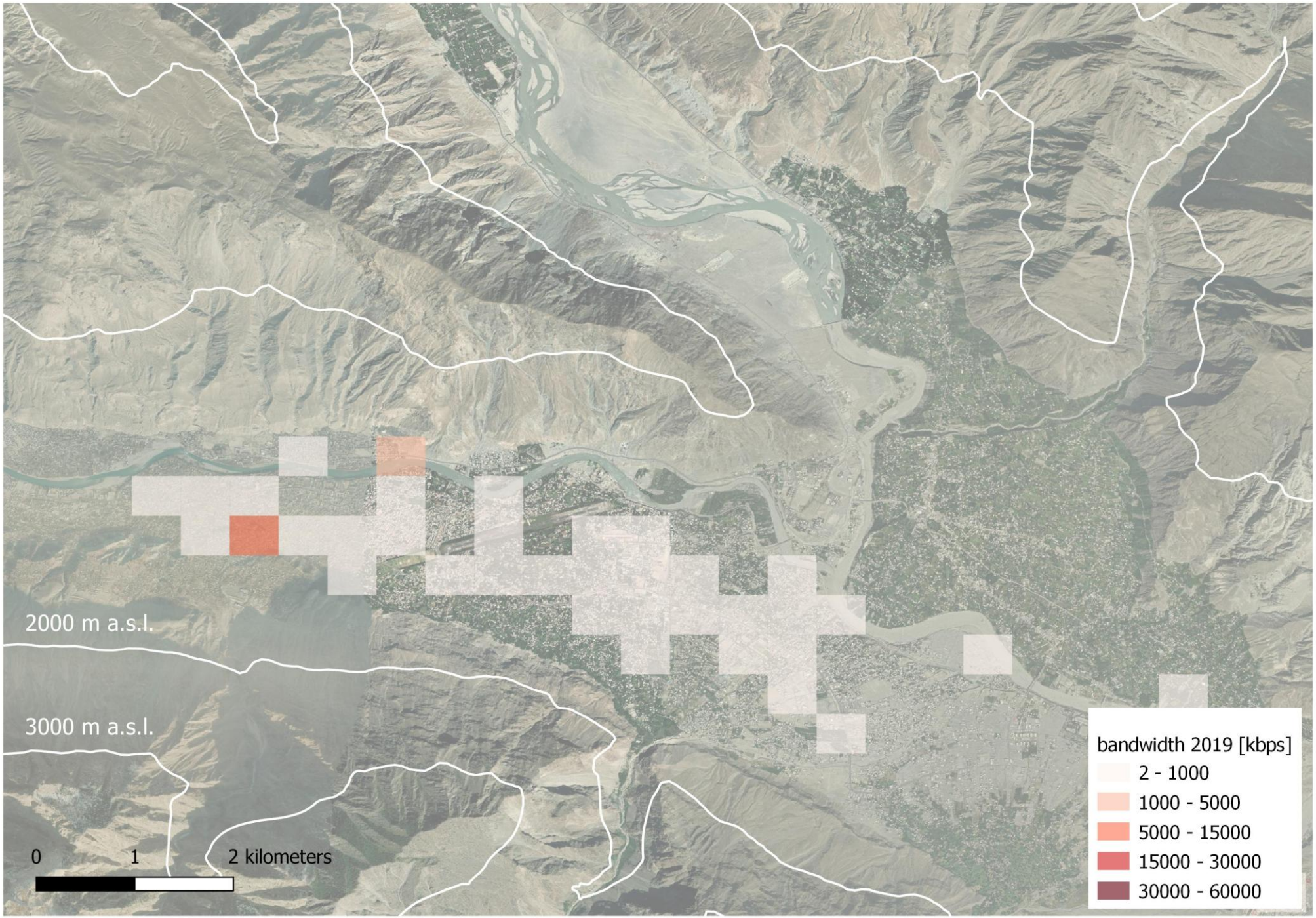
Josias Láng-Ritter, Marko Keskinen & Henrikki Tenkanen

*Nature Communications* **16**, Article number: 2170 (2025) | Cite this article

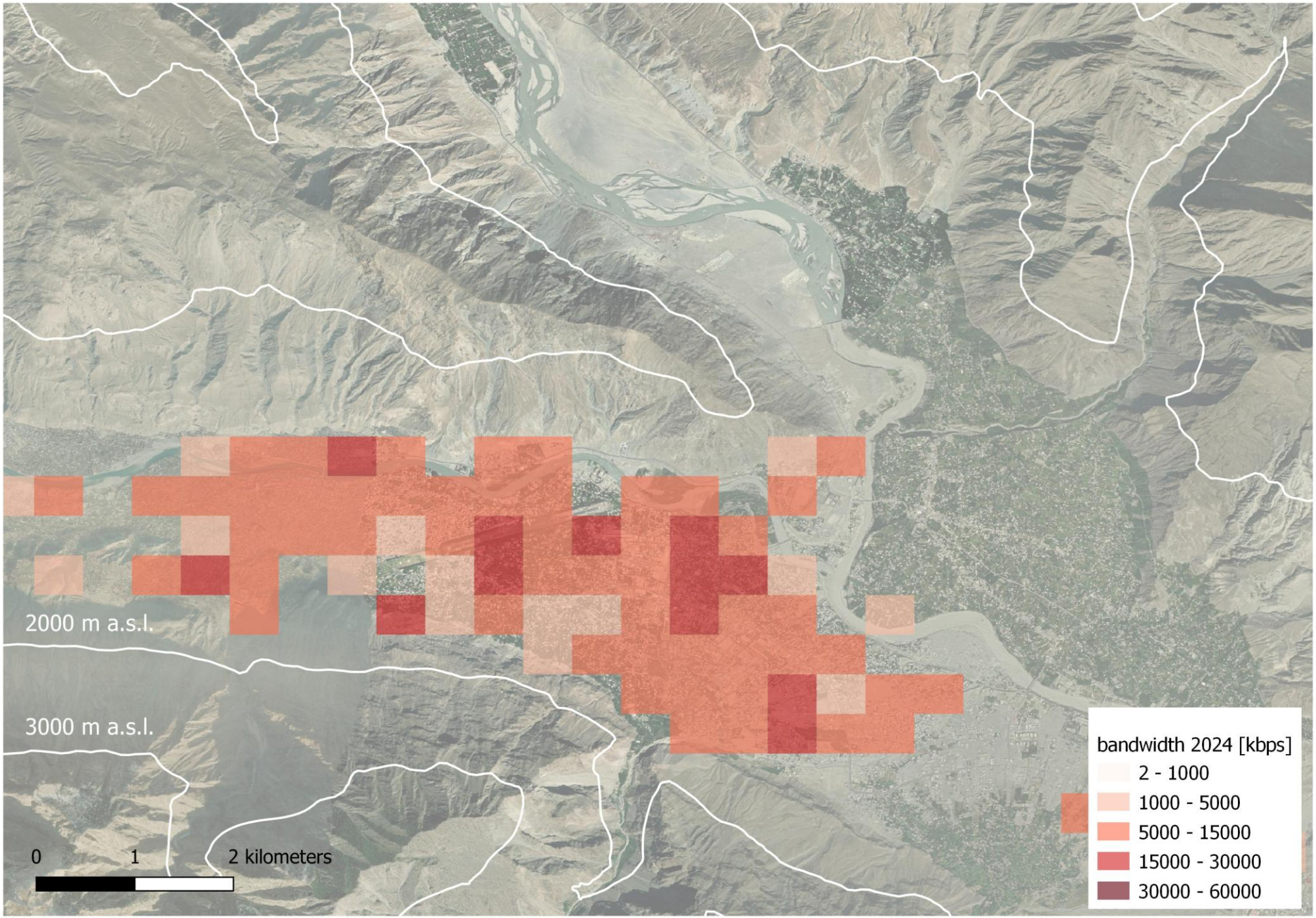
WorldPop Population
data using nightlights
(2020)




Meta (facebook) Population data usin machine learning/ optical data (2020)




Internet connectivity
Ookla 2019




Internet connectivity
Ookla 2024







myRepublica (2021)



# Course content

**22 May (1400 – 1700)**

- 14:00 - 14:30 Introduction to the course and topics
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# General definitions

**Vulnerability**: *the propensity or predisposition to be adversely affected*

**TABLE 1. Evolution of the scientific definition of vulnerability**

| Emphasis on geographic/biophysical context | Emphasis on both geographic context and socio-economic drivers | Emphasis on structural vulnerability; social and economic conditions |
|---|---|---|
| IPCC AR1 and AR2<br>*"the extent to which climate change may damage or harm a system; it depends not only on a system's sensitivity, but also on its ability to adapt to new climatic conditions."* (IPCC, 1997: pg. 504)<br>Focused on hazards and exposure, specifically biophysical and impact assessment models; ability to adapt entrenched in the natural system.<br>outcome vulnerability, equated to total impacts after a climate related event | IPCC AR3 (Third Assessment Report TAR)<br>*"the extent to which a natural or social system is susceptible to sustaining damage from climate change."* (IPCC, 2001: pg. 89)<br>Function of three components : sensitivity, adaptive capacity, and exposure.<br>IPCC AR4<br>*"the degree to which a system is susceptible to, and unable to cope with, adverse effects of climate change, including climate variability and extremes."* (IPCC, 2007: pg. 6)<br>Interacting social and natural systems (socio-ecological contexts); emphasis on resilience<br>Brought climate and development together linked to Special Report on Disasters ("social construction of disasters")<br>multidisciplinary perspectives and increased attention to adaptive capacities (physical in natural systems and social) | IPCC AR5 and AR6<br>*"the propensity or predisposition to be adversely affected and encompasses a variety of concepts and elements, including sensitivity or susceptibility to harm and lack of capacity to cope and adapt."* (IPCC, 2022: pg. 43)<br>Focus on risk, and the socio-economic drivers of vulnerability and development; distinction of structural vulnerability<br>• *contextual* vulnerability that would exist even in the absence of climate change<br>• increased attention to inequity, the role of power, gender, different knowledge systems and values, and bottom up participation in vulnerability assessments<br>• disproportionate effects of climate change and climate justice |

Anisimov and Vallejo(2023)

# Vulnerability in a legal context

UNFCCC (*UNITED NATIONS FRAMEWORK CONVENTION ON CLIMATE CHANGE - https://unfccc.int/resource/docs/convkp/conveng.pdf*) refers to vulnerable countries as:

"low-lying and other small island countries, countries with low-lying coastal, arid and semi-arid areas or areas liable to floods, drought and desertification, and developing countries with **fragile mountainous ecosystems**" (UNFCCC, 1992: 2).

In terms of funding, which includes insurance and the transfer of technology, Article 4.8 outlines that developed country Parties shall consider the "specific needs and concerns" of developing country Parties in the face of climate change and lists the following: "(a) Small island countries; (b) Countries with low-lying coastal areas; **(c) Countries with arid and semi-arid areas, forested areas and areas liable to forest decay; (d) Countries with areas prone to natural disasters; (e) Countries with areas liable to drought and desertification; (f) Countries with areas of high urban atmospheric pollution; (g) Countries with areas with fragile ecosystems, including mountainous ecosystems.**"

# Multidimensional vulnerability index

- **Economic vulnerability** – the risk of harm from exposure to adverse external economic shocks.
- **Environmental vulnerability** - the risk of harm from exposure to natural hazards. Natural hazards may result from structural vulnerability to climate change and anthropogenic or socio-natural shocks and stressors that are exogenous in origin.
- **Social vulnerability** – the risk of harm from exposure to social shocks.
- **Structural economic resilience** – the inherent economic capabilities and economic capital of a country that strengthen the economy's ability to withstand and recover from adverse events.
- **Structural environmental resilience** – the inherent environmental capital of a country, including the ecological resources, infrastructure, and systems that contribute to reducing vulnerability to environmental shocks and stressors.
- **Structural social resilience** – the inherent social capabilities and social capital within a country including social cohesion, social institutions, demographic structure, and human capital that enhance the capacity to withstand and adapt to shocks and stressors.

# Vulnerability in real life scenarios

Lack of education

Lack of basic health care

Lack of savings

Lack of rule of law/governance

# General definitions

**Adaptation**: *to take action to prepare for adverse effects (e.g. due to climate change)*

*IPCC definition*

In human systems, the process of adjustment to actual or expected climate and its effects, in order to moderate harm or exploit beneficial opportunities.

In natural systems, the process of adjustment to actual climate and its effects; human intervention may facilitate adjustment to expected climate and its effects.

**Mal-adaptation**: *when actions to adapt to one outcome, negatively affect the ability to do so elsewhere*

# General definitions

**Mitigation**: making impacts (of climate change) less severe by preventing or reducing the source of adverse drivers (e.g. emission of greenhouse gases (GHG) into the atmosphere)

**Resilience:** Could be translated as the opposite of vulnerability – the ability to withstand or bounce back.

# General definitions

**loss and damage and Loss and Damage**:

- Lower case: simply describes the fact that adverse climate impacts cause irreparable loss and damage to infrastructure, livelihoods etc. and that these can be quantified
- Upper case: the international and political process and discussion around 'who pays for what loss and damage', ongoing since 1991 during the development of the UNFCCC and largely unresolved

- More difficult to assess in general in regions where data is scarce and/or fractured (like in mountains)

# General definitions

Disaster Risk Reduction

*Disaster risk reduction is aimed at preventing new and reducing existing disaster risk and managing residual risk, all of which contribute to strengthening resilience and therefore to the achievement of sustainable development. (UNDRR)*

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# Breakout discussions

In your view and based on your experience

- what are likely the most pressing issues that are being discussed at the global stage, when it comes to addressing risks in mountain regions? This can be climate related or with respect to socio-economic challenges.

- thinking about the future, which risks do you think will change or may only appear in future with our knowledge on anticipated climatic and socioeconomic changes?

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# Literature


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